VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting

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Abstract

Unlearning methods for large language models honour deletion requests with no statistical certificate that forgetting occurred. Existing evaluations are fixed-sample tests: unsound without randomised inclusion, void under optional stopping, and reliant on retraining. We introduce *VOUCH*, an anytime-valid certification framework. At fine-tuning time the provider plants *paired ghost canaries*, exchangeable twins: a fair coin admits one into training and later into the forget set; the other is never trained on. Under exact unlearning the twins are interchangeable, so their score difference is symmetric *by construction*: an exact, distribution-free, finite-sample null. Betting e-processes yield a confidence sequence on the residual advantage, a two-sided (ε, α)-forgetting certificate by sequential equivalence testing, and a revocation alarm. All canaries share one trained model, so their signs are dependent; we certify the *realised advantage on the committed cohort* against a without-replacement boundary, exact under arbitrary dependence, where a fixed boundary breaches its level eightfold. Certificates compose by consensus, alarms by product, and the adaptive-mixture alternative of our own earlier design is anytime-invalid, falsely certifying **99.6%** of the time under an adversarially constructed composite null. Against four valid sequential comparators and on TOFU and MUSE-News on three architectures, VOUCH certifies exact unlearning down to $\varepsilon = 0.05$ on the cohort target, revokes un-unlearned models within tens of pairs, and exposes gradient ascent that destroys the model, at $2Qm$ forward passes. The certificate concerns a declared score class on a committed cohort under an honest-but-verifiable provider, and we measure what each qualification costs.

Keywords: machine unlearning, anytime-valid inference, e-values, membership inference, certification, large language models

1 Introduction

When a user invokes the right to erasure, a model provider must do more than run an unlearning routine: it must be able to *show*, to a regulator or an internal audit function, that the routine worked. This obligation is concrete, though it is worth locating it precisely. Article 17 of the GDPR creates the right to erasure itself, but the duty to *demonstrate* rather than merely assert compliance lives in Articles 5(2) and 24(1), which require controllers to be able to show that processing conforms to the Regulation, and in Article 58(1)(b), which empowers supervisory authorities to conduct data-protection audits [1]. The EU AI Act adds obligations of a different shape: automatic event logging over a system's lifetime (Article 12), technical documentation and its ten-year retention (Articles 11 and 18), production of that documentation and those logs to authorities on reasoned request (Article 21), conformity assessment (Article 43), and the AI Office's power to evaluate general-purpose models (Article 92) [2]. Neither instrument prescribes a statistical test for erasure. Both presuppose that a provider can produce evidence, and that is the gap this paper addresses.

The algorithmic side of machine unlearning has risen to meet the demand: gradient ascent and its stabilised successors, negative preference optimisation [3, 4], representation misdirection [5], and task-vector negation now remove targeted knowledge from large models at modest cost. The *verification* side has not kept pace, and a substantial recent literature documents why: forget-quality metrics track surface behaviour rather than retained knowledge and collapse under attack [6], output-matching scores hide degenerate models [7], benign relearning on unrelated public text restores supposedly erased content [8], and gradient-ascent objectives redistribute probability mass onto paraphrases so that standard metrics report success while the information remains [9, 10]. Evaluation, not optimisation, is the bottleneck.

Three obstacles.

Three specific obstacles stand between current practice and a defensible certificate. *First*, a high membership-inference score on a supposedly forgotten example is not evidence that the example was ever trained on. Distribution shift and the model's own priors confound the signal, and Zhang et al. [11] have shown formally that membership cannot be established without controlling the randomness of what enters training. *Second*, deletion requests arrive as a stream and auditors inspect the evidence as it accrues, stopping when they are satisfied; every fixed-sample test in current use—the forget-quality metric of Maini et al. [12], the leakage metric of Shi et al. [13], the shadow-model panels of Hayes et al. [14]—has its error guarantee voided by exactly this optional stopping, and we measure the damage in Section 5.2. *Third*, the comparators these tests rely on—a model retrained from scratch without the deleted data—are prohibitively expensive at the scale of modern language models and, as Yu et al. [15] show, are not even well defined for the multi-stage pipelines used in practice, because path-dependence means no unique retrained model exists to compare against.

The design.

VOUCH addresses all three with a single design decision and a single inferential engine. The design decision is the *paired ghost canary*. At fine-tuning time we generate exchangeable twin records, and for each pair we flip a fair coin to decide which twin is admitted into training; the admitted twin is later appended to the forget set, so that it passes through the very same unlearning pipeline as organic data, while its twin is withheld from every model. This construction is what makes the audit sound: under exact unlearning the model is independent of the coin, so the two twins are exchangeable and the difference between their scores is symmetric about zero *regardless of the model, the scoring function, or the data distribution*. We therefore obtain an exact, distribution-free, finite-sample null that we generated ourselves, rather than one we must assume. The engine is the betting e-process from game-theoretic statistics, which turns this null into a test whose validity holds not at one predetermined sample size but at every stopping time at once, by Ville's inequality. Streaming deletions and continuous public monitoring, fatal to a fixed-sample test, are precisely what an e-process is built to survive.

What the certificate claims, and what it does not.

We state the scope here because it governs how every later number should be read, and because a certificate whose scope is overclaimed is worse than none. A VOUCH certificate asserts that the residual advantage extractable by a *declared class $\mathcal{F}$ of scores*, on the *committed canary cohort*, against the *deployed model*, is below a declared tolerance ε, under an *honest-but-verifiable* provider who genuinely executes the unlearning routine on the forget set as declared. Four qualifications follow, and each is quantified rather than merely disclosed. The guarantee is per-cohort, not per-deletion-request, since per-request certificates would need per-user canaries. It concerns $\mathcal{F}$, and that is a definitional limit: two weak attacks from outside $\mathcal{F}$ (Section 5.13) do *not* beat the noise floor of subsampling and so settle nothing either way, while a LiRA-style shadow-model attack (Section 5.14)—demonstrably potent, recovering an advantage of 0.82 on an un-unlearned model—finds none on a certified one, which bounds the gap on that tier without closing it in general. The transfer from the realised cohort to the mechanism-level advantage costs an explicit inflation of $\kappa\sqrt{2\log(2/\alpha')/m}$ (Proposition 3), which is why our tight-tolerance results are cohort statements. And the honest-but-verifiable assumption is load-bearing: Section 5.16 shows a zero-knowledge perplexity filter recovers the entire planted cohort, so a provider who wished to special-case it could find it, and only cryptographic attestation [16] closes that gap. What detection does not do is defeat the audit, because both twins in a pair are equally conspicuous and locating the cohort reveals nothing about which twin the coin admitted.

Corrections carried by this version.

Building the framework and running it end-to-end forced five corrections on the initial design, and because each turned out to matter we foreground them rather than bury them. *(i)* The certificate's null must be stated conditionally, not marginally. Signs from all canaries pass through one jointly trained model, so they are dependent; a fixed-boundary process that assumes otherwise is breached eightfold under model-level over-dispersion (Table 1). We retarget the certificate at the realised cohort advantage and bet against a without-replacement boundary, which is exact under arbitrary dependence (Theorem 2). *(ii)* The natural way to combine several attack scores into one certificate—bet on an adaptively weighted mixture of their signs—is *anytime-invalid*, and in simulation our own earlier design falsely certified a leaking model 99.6% of the time under an adversarially constructed composite null; the sound alternative requires every score's own process to concur. *(iii)* Certificates and alarms must compose in *opposite* directions across deletion waves, and multiplying certificate e-values, the instinctive move, certifies histories that contain a genuinely bad wave. *(iv)* Gradient-ascent unlearning frequently *over-forgets* on real language models, driving the forgotten twins to scores conspicuously *below* their never-seen partners—membership leakage that a one-sided test happily certifies and that a two-sided test must catch. *(v)* The small-ε expansion of the Kullback–Leibler drift is $\varepsilon^2/2$, not $\varepsilon^2/8$, so the cohort-size rule is $2\log(1/\alpha)/\varepsilon^2$; the numerical columns were always computed from the exact divergence and are unaffected, but the stated rule was wrong by a factor of four.

Contributions.

We formulate unlearning certification as sequential two-sided equivalence testing against a declared attack class, with an exact distribution-free null supplied by paired ghost canaries (Section 3). We give the supporting theory: an exact conditional null requiring no independence across pairs, a certificate that is exact under arbitrary dependence via without-replacement betting, a priced bridge to the super-population advantage, sound composition across scores and across deletion waves in both directions, power that matches the Kullback–Leibler information limit, an exact finite-cohort certification-time bound that holds for every betting strategy and every reveal order, and a one-directional consistency bridge from certified $(\varepsilon_u, \delta_u)$-unlearning (Section 4). We report a comprehensive empirical study (Section 5): calibration under adversarial optional stopping over 2,000 seeds; head-to-head comparison against four *valid* sequential procedures; end-to-end certification of gradient ascent, GradDiff, NPO and retraining on TOFU and MUSE-News across three architectures, with SimNPO and representation misdirection added on TOFU/GPT-2 and the latter confirmed on a second architecture, a tolerance sweep down to $\varepsilon = 0.05$, utility guardrails, and dose–response calibration; a head-to-head against the descriptive metrics VOUCH is argued to replace; attacks from outside the declared score class, including a LiRA-style shadow-model attack with matched shadows and a positive control; a paraphrase-aware score added to the declared class and priced; a measurement of how detectable the planted cohort is, with an entropy-dilution design rule that follows from it; recoverability probes that relearn, quantise, and jailbreak the unlearned model; a model axis spanning 2019 to 2026; and cost measurements with their hardware and workload assumptions stated. All code, all results, and the fault-tolerant harness that produced them on free ephemeral compute are released.

2 Related work

Our work sits at the meeting point of four literatures—unlearning algorithms, the measurement of memorisation and forgetting, the verification of unlearning, and game-theoretic statistics—and it is the gap between the middle two that game-theoretic statistics turns out to close. We survey each in turn and then state precisely how VOUCH differs from its nearest neighbours.

2.1 Unlearning algorithms and their benchmarks

Exact unlearning by architectural design, exemplified by sharded training [17], retrains only the shards touched by a deletion; it is sound but cannot be applied to a model that was not trained for it, which is the situation every deployed language model is in. The practical response has therefore been *approximate* unlearning, and for large language models a compact toolbox has emerged: gradient ascent on the forget set and its retain-regularised variant GradDiff; negative preference optimisation [3], which replaces the unbounded ascent objective with a bounded preference loss and which its authors introduced precisely to avoid the catastrophic collapse that ascent invites; the simplified reappraisal SimNPO [4], which discards NPO's frozen reference model in favour of a length-normalised margin objective and which we include as a subject in Section 5.8; representation misdirection [5], which acts on hidden activations rather than on the token loss and which we add as a subject in Section 5.9; and task-vector negation. These methods are evaluated on purpose-built benchmarks—TOFU's fictitious-author question answering [12], MUSE's six-way news-and-books evaluation [13], and the hazardous-knowledge multiple-choice of WMDP [5]—and we treat all of them, methods and benchmarks alike, as *subjects* that VOUCH certifies rather than as competitors.

2.2 Measuring memorisation and forgetting

The canary methodology VOUCH depends on descends directly from the secret-sharer line of work. Carlini et al. [18] insert random secret sequences into training data and define *exposure*, the canary's rank among all possible instantiations ordered by model log-perplexity, converting memorisation into a calibrated number; they show that memorisation is commonplace, that it is not explained by overfitting, and that it is eliminated only by differentially private training. VOUCH borrows the planted-canary device and changes one thing that turns out to matter for inference: where the secret sharer plants a canary and *measures* its exposure, we plant a *pair* and randomise which twin enters training, so that the measurement acquires a null. Exposure answers "how memorised is this string"; the ghost twin answers "could this have happened by chance," which is the question a certificate must answer and the one Zhang et al. [11] show cannot be answered without controlled randomisation.

The strongest membership attacks in this lineage calibrate per example rather than thresholding a raw score. Carlini et al. [19] show that comparing a target model's confidence on an example against the confidences that shadow models trained with and without it produce—a likelihood-ratio test they call LiRA—dominates score-threshold attacks, particularly in the low false-positive regime that matters for a privacy claim.

LiRA is the natural adversary against which to stress-test a declared score class, since it uses information no member of ours has: N additionally trained models. It is for the same reason not a candidate for inclusion in $\mathcal{F}$, because a verifier held to $2Qm$ forward passes and no training cannot run it. We use it in Section 5.14 as an attack from outside the class, with a positive control that establishes it is working before its null on a certified model is read as meaning anything.

The complementary literature measures forgetting rather than memorisation. Jagielski et al. [20] quantify how the success of privacy attacks on a training example decays as training continues past it, and find that standard models do forget over time—but also prove that non-convex models can retain memorised data indefinitely, and show that forgetting disappears entirely under deterministic training. Their conclusion is the premise of ours: incidental forgetting is real but uncontrolled, and therefore cannot substitute for a certificate. Bărbulescu and Triantafillou [21] document, in the same spirit, how per-sequence memorisation strength governs how hard a datum is to remove, which is the phenomenon our repetition strata are designed to traverse.

2.3 Evaluating unlearning: a literature of negative results

What the unlearning literature has increasingly documented is that the metrics used to declare success are unreliable, and this body of work is the direct motivation for VOUCH. A first strand shows that apparent forgetting is often mere suppression. Łucki et al. [22] recover ostensibly unlearned capabilities through lightweight attacks; Zhang et al. [23] show that quantising an unlearned model to 4 bits resurrects forgotten knowledge; Deeb and Roger [24] recover most pre-unlearning accuracy by fine-tuning on adjacent facts; and Fan et al. [25] formalise relearning attacks and connect robustness to sharpness-aware minimisation. Closest to our P1 probe is Hu et al. [8], who show that fine-tuning an unlearned model on a small, *benign*, publicly available corpus that never contains the forget targets is enough to jog the model's memory and restore both hazardous knowledge and verbatim memorised passages. Their framing—obfuscation versus removal—is precisely the distinction our relearn probe operationalises, and their result is why we treat a post-relearning verdict as part of the certificate rather than as an optional extra: on our benchmark tier at $\varepsilon = 0.2$ the same probe revokes NPO's certificate on one MUSE/GPT-2 seed and one MUSE/Phi-1.5 seed, and leaves it intact on every TOFU cell (Section 5.8).

A second strand indicts the evaluation protocols themselves. Feng et al. [10] demonstrate that current LLM-unlearning evaluations can be made to both overstate and understate success. Wang et al. [6] show that forget-quality metrics track surface output behaviour rather than retained knowledge and collapse under red-teaming, and—more damagingly for cross-paper comparison—that the reported unlearning/retention trade-off does not lie on a Pareto frontier, so method rankings are artifacts of hyperparameter choice; they propose recalibrating retained performance before comparing. Yuan et al. [7] make the complementary point that ROUGE-style output matching hides degenerate behaviour, and add metrics for token diversity, sentence semantics, and factual correctness. Both diagnoses apply to VOUCH's subjects and shaped how we report them: we do not rank unlearning methods against one another, we certify each one against a declared tolerance and pair every verdict with a utility reading, precisely because a ranking built on a non-Pareto trade-off would not survive a change of hyperparameters. Our own worked example in Figure 5 is an instance of the degeneracy Yuan et al. [7] warn about—an NPO-unlearned model emitting `LLLLLL...` on canary prompts while its held-out loss is untouched—and it is caught here not by a fluency metric but by the two-sided certificate, which treats scoring conspicuously *below* the ghost twin as leakage.

A third strand is mechanistic and, for our purposes, the most pointed. Wei et al. [26] show that facts presumed forgotten persist through correlated knowledge, and Xu et al. [27] distinguish genuine erasure from suppression at the representation level, finding behaviour easily restored by minimal fine-tuning. Li et al. [9] identify the mechanism they call the *squeezing effect*: gradient-ascent-style objectives push probability mass off the target response but redistribute it onto nearby high-likelihood regions, typically semantically equivalent rephrasings, so the content resurfaces under paraphrase and commonly used metrics report success while semantically related information remains. Their remedy couples the unlearning objective to the model's own high-confidence generations. This result bears directly on the scope of what VOUCH certifies and we want to be exact about the relationship. VOUCH is a *verification* framework and inherits the limits of its declared score class: our scores read the likelihood of the literal secret span under a fixed set of query wrappers, so probability mass that has migrated to a paraphrase is, by construction, mass our default class does not see. The framework's response is structural rather than rhetorical—the class $\mathcal{F}$ is declared, the certificate is explicitly a statement about $\mathcal{F}$, and any score a sceptic can name can be added to $\mathcal{F}$ at the cost of a slower certificate and no loss of validity, since Theorem 2 needs no multiplicity correction. Sections 5.13 and 5.14 run attacks from *outside* the declared class against already-certified models and report where they exceed the tolerance. We also implement the score their result calls for: Section 5.15 adds a paraphrase-aware member to $\mathcal{F}$, which costs one to two per cent in pairs and changes no verdict—though we are explicit there that our random-string secrets have no semantic neighbourhood for mass to migrate into, so this tests the score's cost rather than the squeezing effect itself, which would need canaries whose secret is paraphrasable.

The recurring message across all of this is that *evaluation*, not optimisation, is the bottleneck—which is exactly the problem VOUCH addresses, and which motivates our recoverability probes (R-VOUCH) and our insistence on pairing every certificate with a utility reading.

2.4 Verifying and certifying unlearning

The earliest guarantees came from *certified removal* [28–31], which bounds the distance between the unlearned and retrained models in a differential- privacy-style $(\varepsilon_u, \delta_u)$ sense—the same framing Dang [32] argues is the more defensible reading of "forgotten" than the erasure language of the GDPR itself, since it requires that no test can tell whether the deleted record was ever present, exactly the guarantee a ghost-canary pair is built to supply on the model VOUCH certifies. These results are elegant but confined to convex or near-convex regimes, and their bounds become vacuous at the scale of modern language models; we use them not as a baseline but as a consistency check, showing in Proposition 8 that any $(\varepsilon_u, \delta_u)$-certified algorithm passes VOUCH—a one-directional implication, as Remark 4 states. The verification of *arbitrary* unlearning has proven far harder. Thudi et al. [33] argue that approximate unlearning cannot be audited from weights alone and that auditable definitions are a prerequisite for any guarantee; Zhang et al. [34] show that a dishonest provider can defeat both existing families of verification strategies while passing them; and, most consequentially for our design, Zhang et al. [11] establish that membership-inference scores *cannot* prove that a model was trained on given data unless the inclusion of that data was randomised—the soundness principle that our fair-coin canary design is built to satisfy. Two further impossibility results bound what any behavioural certificate can claim: Yu et al. [15] show that path-dependence makes retrain equivalence unattainable for multi-stage pipelines, which is why VOUCH's null is internally generated rather than referenced to a retrained model, and the reversibility findings above imply that no behavioural method can certify parameter-space erasure—a limit we state explicitly rather than paper over.

A handful of recent efforts move toward statistical certification, and these are our closest neighbours. Pandey et al. [35] give a hypothesis-testing notion of certified unlearning, but it is fixed-sample and tied to a Gaussian high-dimensional regression regime rather than to LLM-scale behavioural auditing. Pawelczyk et al. [36] use data-poisoning as a lens on unlearning failure but provide no error-controlled certificate. Cryptographic proof-of-unlearning [16] attests that the correct computation *ran*, which is orthogonal to whether forgetting *holds* statistically and with which VOUCH composes—and which is the right tool against the special-casing provider our threat model excludes. The recent survey of Xue et al. [37] organises this fast-moving area and confirms that no existing method delivers what VOUCH does: an anytime-valid, distribution-free certificate of bounded residual influence. The broader programme is surveyed by Liu et al. [38].

2.5 Game-theoretic statistics

The inferential engine we borrow is that of safe, anytime-valid inference: e-values, test martingales, and betting confidence sequences [39–43]. The defining property, traceable to Ville's inequality [44], is that a nonnegative martingale started at one exceeds $1/\alpha$ with probability at most α *at every stopping time simultaneously*; this is precisely what a streaming, continuously-monitored deletion audit requires, and precisely what a fixed-sample test cannot offer. The betting perspective [42, 45] additionally supplies confidence sequences whose bets adapt online to near-optimal growth, and—the ingredient our corrected certificate turns on—the without-replacement construction for finite-population means, in which the null boundary is recentred at each step on the mean the unsampled remainder would need to carry. That device is what lets Theorem 2 dispense with independence across pairs entirely. The one adjacent application of these tools is to privacy auditing: Steinke et al. [46] audit differential privacy in a single training run, and González et al. [47] and Csillag and Mesquita [48] bring sequential e-value machinery to DP auditing. The distinction is fundamental to the hypothesis structure. Privacy auditing *lower*-bounds the privacy loss of a *training* algorithm by mounting a successful attack; VOUCH *upper*-bounds the residual influence *after unlearning* by showing that no attack in a declared class succeeds—an equivalence test, not a significance test, run over ghost twins that have passed through the unlearning pipeline. To our knowledge, VOUCH is the first application of game-theoretic statistics to unlearning certification.

2.6 Positioning

The contrast with our nearest neighbours can be summarised sharply. Fixed-sample paired or two-sample tests—including the KS-based forget-quality metric of Maini et al. [12] and the min-$k\%$ leakage metric of Shi et al. [13]—are invalidated by optional stopping and typically require a retrained reference; Section 5.12 runs both on our own models and seeds and reports where they and VOUCH disagree. The sound-membership results of Zhang et al. [11] tell us *why* controlled randomness is indispensable but stop at a fixed-n membership proof rather than a sequential forgetting certificate; the secret-sharer and forgetting-measurement lines [18, 20] supply the canary device and the decay measurements but no null and no error control; shadow-model evaluators [14] are descriptive and cost dozens of fine-tunes; certified removal [28, 29] does not scale; and DP auditing [46, 47] solves the opposite, lower-bounding, problem. VOUCH is, to our knowledge, the first framework to formulate unlearning certification as sequential equivalence testing and to deliver exact, finite-sample, anytime-valid certificates of bounded residual influence at LLM scale, with soundness inherited from a randomised paired-canary design rather than from distributional assumptions or shadow models.

3 Setting and the VOUCH protocol

This section develops the protocol in full: the actors and threat model, the canary construction that manufactures the null, the filtration that makes the martingale argument literally correct, the certified quantity, the betting engine with its explicit wealth dynamics, and finally the assembled protocol as pseudocode (Algorithm 1) and as a flowchart (Figure 1).

3.1 Actors, pipeline, and threat model

The provider fine-tunes a base model on $D_{\text{keep}} \cup D_{\text{forget}} \cup C$, where C is the canary set described below. When deletion requests arrive, an unlearning algorithm $\mathcal{U}$—negative preference optimisation, GradDiff, and so on—produces an unlearned model M_u from the fine-tuned model, using the forget set augmented with the relevant canaries. A verifier, holding only black-box query access to M_u together with a canary manifest whose commitment was published before unlearning began, runs VOUCH and either issues, withholds, or revokes a certificate.

Throughout we assume the *honest-but-verifiable* setting: the provider genuinely executes $\mathcal{U}$ on the forget set as declared, and the open question is whether it worked and how well, under quantified error control. This assumption is doing real work and we state it wherever the certification claim is summarised, because the commitment prevents the manifest from being changed after the fact but does not prevent a provider who has *located* the cohort from treating it differently from organic data. Section 5.16 measures how hard locating the cohort actually is—the answer is that it is cheap—and explains why locating it is nonetheless not the same as defeating the audit: both twins in a pair are equally conspicuous, so a provider who finds the cohort still learns nothing about which twin the coin admitted. A provider willing to special-case the committed canaries is outside our scope and, as prior work establishes, can only be addressed by cryptographic attestation [16], with which VOUCH composes.

3.2 Paired ghost canaries

A templated generator emits exchangeable twin pairs (c_i^0, c_i^1) for $i = 1, \dots, m$. The two twins in a pair are independent draws from the same conditional template distribution—the same surface form populated with independently sampled fictitious entities carrying secrets of at least 40 bits of entropy—so that within a pair they are exchangeable by construction. We match the template to the host corpus, using question–answer pairs for TOFU and record-style sentences for news text. For each pair an independent fair coin $b_i \sim \mathrm{Bern}(\frac{1}{2})$ selects the *in-twin* $c_i^{b_i}$, which we insert into the fine-tuning corpus with a repetition factor $r_i \in \{1, 2, 4, 8\}$ (these strata give us a dose–response curve) and later append to the forget set, so that it traverses the unlearning pipeline exactly as organic forget data does. The *ghost twin* $c_i^{1-b_i}$ is never shown to any model.

3.3 Commitment and filtration

Before unlearning begins the provider publishes a commitment to the manifest. The commitment is a Merkle tree whose i-th leaf is

$$\ell_i \;=\; H\big(c_i^0,\, c_i^1,\, b_i,\, r_i,\, \rho_i\big), \qquad H = \text{SHA-256}, \tag{1}$$

with ρ_i a per-leaf blinding nonce derived from the manifest seed, and the published value is the tree root. This binds the pairs and the coins and prevents the audit from being gamed after the fact, exactly as a flat hash of the whole manifest would; the tree structure buys one further thing that a flat hash does not, and the theory needs it. At step t the provider hands over leaf $\pi(t)$ together with its authentication path, which the verifier checks against the published root; the path is $\lceil \log_2 m \rceil$ hashes long, so per-pair opening costs 9 hashes at $m = 384$. The blinding nonce matters because the template library is public: without it, an unopened leaf's hash could be attacked by enumeration.

The verifier draws a uniformly random reveal order π, committed in advance, and opens leaf $\pi(t)$ at step t, checking its Merkle path against the published root. Where π's randomness comes from is a protocol requirement rather than an implementation choice, because Theorem 2 spends it: it must be the verifier's own, drawn after the commitment and unpredictable to the provider (Remark 1). The filtration is

$$\mathcal{F}_t \;=\; \sigma\Big(\text{query access to } M_u,\; \big\{(c^0_{\pi(u)}, c^1_{\pi(u)}, b_{\pi(u)}, r_{\pi(u)})\big\}_{u \le t},\; \text{tie coins } 1..t\Big). \tag{2}$$

Opening leaf by leaf rather than all at once is what makes $b_{\pi(t)}$ genuinely unrevealed at time $t-1$, and hence what makes "the coin is still fair given the past" a true statement about $\mathcal{F}_{t-1}$ rather than an informal one. Every supermartingale argument in Section 4 is with respect to (2). Nothing else about the protocol changes, and the verifier's computational cost is unaffected.

3.4 Scores and the certified quantity

We fix a declared class $\mathcal{F}$ of scalar score functions, each aggregated over Q fixed query prompts, and each oriented so that a *higher* score means *more memorised*:

- s_{loss}, the token-normalised *log-likelihood* of the secret span—that is, the negative of its token-normalised negative log-likelihood;
- s_{mink}, the mean of the lowest $k\%$ token log-probabilities of the span ($k = 20$);
- s_{ratio}, the base-model-calibrated difference $s_{\text{loss}}(M_u, \cdot) - s_{\text{loss}}(M_0, \cdot)$, available when the verifier may query the pristine base model;
- optionally s_{probe}, a linear probe on the hidden state, calibrated on a disjoint cohort.

The sign convention matters and we fix it once here, since a log-likelihood and a negative log-likelihood differ by exactly the sign of every reported gap. For pair i and score s we form the within-pair difference and its sign,

$$D_i^{(s)} = s(M_u, c_i^{\text{in}}) - s(M_u, c_i^{\text{ghost}}), \qquad Z_i^{(s)} = \mathbf{1}\{D_i^{(s)} > 0\}, \tag{3}$$

breaking ties by a committed fair coin. With scores oriented as above, a *positive* D means the trained twin is the better-remembered one, which is the direction of leakage; the worked example of Figure 5 reports $D = +8.0$ for a fine-tuned model whose in-secret costs 0.99 nats per token against the ghost's 9.03, and this is now consistent with (3), with the figure, with the tables, and with the code. Exact ties are rare but not always negligible: we measure a rate of 0 in 124,416 scored pairs for s_{loss} across every architecture, and 2.7% for s_{mink} on Gemma-4-E2B, where the min-$k\%$ statistic over a short span takes few distinct values (Section 5.8). The committed tie-break coin is what keeps the null exact in that cell.

The quantity we certify is $\sup_{s\in\mathcal{F}} |\Delta^{(s)}|$ with $\Delta^{(s)}$ the realised cohort advantage of Definition 1. We are deliberate that this is influence extractable *through* $\mathcal{F}$ on the *committed canary cohort*: it is the strongest target compatible with the impossibility results for behavioural audits, and we make no claim about erasure in parameter space, about scores outside $\mathcal{F}$, or—without the explicit inflation of Proposition 3—about the super-population advantage.

3.5 The betting engine

All three of VOUCH's outputs are built from one primitive, the *betting e-process*, and it is worth writing its dynamics out explicitly because the validity proofs in Section 4 reduce to inspecting them. A gambler starts with unit wealth and, before the pair opened at step t is revealed, stakes a predictable fraction λ_t (chosen from the information $\mathcal{F}_{t-1}$ revealed so far) on the sign $Z_{\pi(t)}$ deviating from a hypothesised mean.

For the *certificate* arm the hypothesised mean is not a constant but the predictable without-replacement boundary $p_0(t)$ of Eq. (11): the mean the unrevealed remainder of the cohort would have to carry for the null $\Delta^{(s)} \geq \varepsilon$ still to hold. Against that boundary and its mirror the wealth processes for score s are

$$E_t^{s,+} = \prod_{u \leq t} \big(1 + \lambda_u^{s,+} \, (p_0(u) - Z_{\pi(u)}^{(s)})\big), \qquad E_t^{s,-} = \prod_{u \leq t} \big(1 + \lambda_u^{s,-} \, (Z_{\pi(u)}^{(s)} - \bar{p}_0(u))\big), \quad (4)$$

with $\bar{p}_0(t)$ the analogous boundary for the null $\Delta^{(s)} \leq -\varepsilon$ and stakes confined to the admissible range so that wealth can never go negative. Because sequential reveal of a uniform permutation is simple random sampling without replacement, the conditional mean of the next sign is exactly the mean of the remainder, so under the null each factor has conditional mean at most one: each process is a nonnegative supermartingale started at one, and Ville's inequality converts wealth into evidence. Wealth above $1/\alpha$ is a licence to reject at whatever time we happen to be looking. Setting $p_0(t) \equiv p_0$ instead recovers the fixed-boundary process of the earlier draft, which targets the super-population advantage and requires the conditional-mean null; we retain it as a reported variant and compare the two throughout Section 5.

For the *revocation* arm the hypothesised mean is the constant $\frac{1}{2}$, which Theorem 1 makes exact conditionally, so no boundary recentring is needed.

How the stakes are chosen determines power, not validity, which frees us to be aggressive. VOUCH plays a uniform mixture of 8 experts and lets their wealth arbitrate: 6 constant-fraction bettors $\lambda = c \cdot \lambda_{\max}$ with $c \in \{0.02, 0.05, 0.1, 0.2, 0.4, 0.7\}$, an online-Newton-step (ONS) learner, and a truncated Krichevsky–Trofimov (KT) plug-in. The ONS expert follows the coin-betting recursion

$$g_t = \frac{p_0(t) - Z_t}{1 + \lambda_t (p_0(t) - Z_t)}, \qquad A_t = A_{t-1} + g_t^2, \qquad \lambda_{t+1} = \Pi_{[0, \lambda_{\max}]}\Big(\lambda_t + \tfrac{\eta \, g_t}{A_t}\Big), \quad (5)$$

with $A_0 = 1$, $\eta = \frac{1}{2}$, and Π the clip onto the admissible range; its log-wealth regret against the best constant bet in hindsight is $O(\log t)$. The KT expert bypasses stake selection altogether: it estimates the mean by $q_t = (\sum_{u<t} Z_u + \frac{1}{2})/t$, truncates q_t to the alternative side of $p_0(t)$, and multiplies wealth by the plug-in likelihood ratio

$$e_t \; = \; Z_t \, \frac{q_t}{p_0(t)} \; + \; (1 - Z_t) \, \frac{1 - q_t}{1 - p_0(t)}, \quad (6)$$

which has conditional mean at most one under the null and achieves the optimal growth rate up to a $\frac{1}{2t}\log t$ redundancy. Because a convex combination of e-processes with a fixed prior is again an e-process, the mixture inherits exact validity while paying only a $\log 8$ additive cost in log-wealth relative to its best expert—power adapts online with nothing to tune.

The same wealth idea, run over a grid rather than at a single boundary, yields the quantitative face of the certificate. For every candidate mean m' on a grid of 1001 points in $[0,1]$ we maintain two wealth processes betting that the truth is above and below m', each recentred by the same without-replacement rule, with empirical-Bernstein (aGRAPA) stakes

$$\lambda_t(m') \;=\; \Pi_{[0,\,\lambda_{\max}]}\Big(\tfrac{\hat{\mu}_{t-1}-m'}{\hat{\sigma}^2_{t-1}+(\hat{\mu}_{t-1}-m')^2}\Big), \tag{7}$$

where $\hat{\mu}_{t-1}$ and $\hat{\sigma}^2_{t-1}$ are the running mean and variance of the signs. The set of grid values whose wealth on both sides stays below $2/\alpha$ and whose recentred boundary remains in $[0,1]$, intersected over time,

$$C_t \;=\; \bigcap_{u\le t}\big\{\, m' :\; \max\big(W^{\uparrow}_u(m'),\, W^{\downarrow}_u(m')\big) < 2/\alpha \,\big\}, \tag{8}$$

is a $(1-\alpha)$ confidence sequence for the cohort sign rate: it covers the truth at every time simultaneously, so the auditor may glance at it whenever they please, and it collapses onto the exact cohort value once the cohort is exhausted. Mapped through $\Delta = 2p-1$ it reports the residual advantage with its anytime-valid uncertainty; Remark 3 states precisely which coverage claims are per-score and which are simultaneous over $\mathcal{F}$.

The revocation arm inverts the role of the null. Under *exact* unlearning, Theorem 1 makes every $Z^{(s)}_{\pi(t)}$ an exact conditional fair coin, so here—and only here—adaptive score weighting is sound. We run, per score and per direction, e-processes against $\mathbb{E}[Z] = \frac{1}{2}$ and combine them with predictable wealth-proportional weights: with $w_{s,t} \propto \exp(\log E^s_{t-1})$ computed before the pair at step t is opened,

$$R_t \;=\; \prod_{u\le t}\sum_{s\in\mathcal{F}} w_{s,u}\,\frac{E^s_u}{E^s_{u-1}}, \tag{9}$$

which is an e-process because each factor is a predictable convex combination of conditional-mean-one increments. When $R_t \ge 1/\alpha$ the verifier holds anytime-valid evidence of residual memorisation and the certificate is withheld or revoked; the weights meanwhile steer wealth toward whichever score discriminates best, which is exactly what an alarm should do.

3.6 The protocol end to end

Algorithm 1 assembles the pieces into the three-phase protocol, and Figure 1 shows the same pipeline as a flowchart. Phase 0 runs at design time: generate the cohort, flip the coins, publish the Merkle root, fine-tune with the in-twins planted. Phase 1

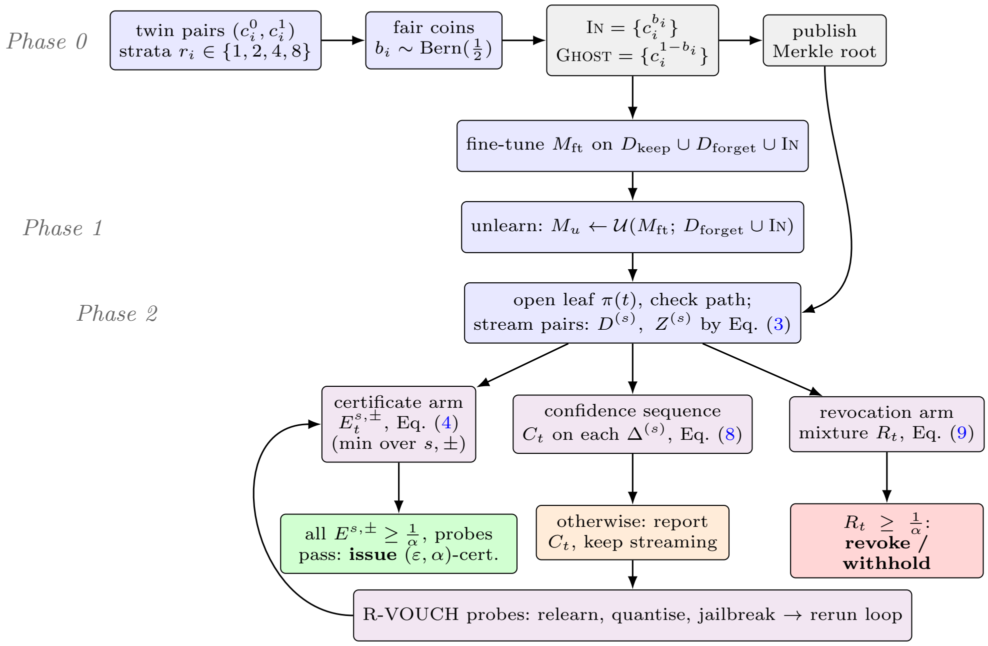

Fig. 1 The VOUCH pipeline. Phase 0 plants paired ghost canaries and commits to the manifest as a Merkle tree; Phase 1 unlearns with the in-twins riding along; Phase 2 opens one leaf at a time in a committed random order and streams the pairs through three coupled anytime-valid processes—the per-score certificate minimum, the betting confidence sequence, and the revocation mixture—whose boundary crossings trigger issuance, continued monitoring, or revocation. R-VOUCH probes rerun the loop on relearned, quantised, and jailbroken variants before a robust certificate is issued.

is the provider's unlearning step, in which the in-twins ride along with the organic forget set. Phase 2 is the audit: stream the pairs in the committed random order, opening and checking one leaf at a time, updating the certificate processes, the confidence sequences, and the revocation mixture after each pair, and acting on whichever boundary is crossed first. R-VOUCH then re-runs the same loop on probed variants of M_u—after relearning on public data, after 4-bit quantisation, and under adversarial prompt wrappers—and the robust certificate requires every probe to pass. Our empirical probe coverage is partial and we state it here rather than leave it to be inferred: P1 and P3 run on every benchmark cell and every seed but are anchored on NPO; P2 runs on the GPT-2 and TinyGPT tiers only, because the shared-frozen-base multi-adapter design would have to materialise merged weights to quantise a multi-billion-parameter model. Streaming deletion waves each draw a fresh cohort with independent coins; Theorem 6 dictates how their verdicts may be combined.

4 Theory

Our guarantees rest on one structural fact—the null we plant is exact—and on the supermartingale property of betting. The proofs are short, and their brevity is the

Algorithm 1 The VOUCH protocol

Phase 0 (design time, before fine-tuning)

generate twin pairs $\{(c_i^0, c_i^1)\}_{i=1}^m$; draw coins $b_i \sim \mathrm{Bern}(\frac{1}{2})$ and strata $r_i \in \{1, 2, 4, 8\}$

$\textsc{In} \leftarrow \{c_i^{b_i}\}$; $\textsc{Ghost} \leftarrow \{c_i^{1-b_i}\}$ ▷ ghosts are never shown to any model

publish the Merkle root over leaves $\ell_i = H(c_i^0, c_i^1, b_i, r_i, \rho_i)$ ▷ Eq. (1)

fine-tune M_{ft} on $D_{\mathrm{keep}} \cup D_{\mathrm{forget}} \cup (\textsc{In}$ with repetitions $r_i)$

Phase 1 (unlearning)

$M_u \leftarrow \mathcal{U}(M_{\mathrm{ft}};\ D_{\mathrm{forget}} \cup \textsc{In})$ ▷ in-twins traverse the pipeline as organic forget data

Phase 2 (certification; black-box queries only)

initialise $E^{s,+} = E^{s,-} = 1$ for all $s \in \mathcal{F}$; revocation mixture $R = 1$; CS grids $C^{(s)}$

for $t = 1, \ldots, m$ in the committed random order π **do**

  open leaf $\ell_{\pi(t)}$; **abort** unless its Merkle path checks against the published root

  for $s \in \mathcal{F}$ **do**

    $D^{(s)} \leftarrow s(M_u, c_{\pi(t)}^{\mathrm{in}}) - s(M_u, c_{\pi(t)}^{\mathrm{ghost}})$; $Z^{(s)} \leftarrow \mathbf{1}\{D^{(s)} > 0\}$ (ties: committed coin)

    recentre the boundary $p_0(t)$ by Eq. (11); update $E^{s,+}, E^{s,-}$ by Eq. (4) with mixture bets (5)–(6); update $C^{(s)}$ by (7)–(8)

  end for

  update the revocation mixture R by Eq. (9)

  if $R \geq 1/\alpha$ **and** not yet revoked **then**

    mark **revoked** at t; **continue** the loop ▷ keep compounding R for the streaming product alarm

  end if

  if not revoked **and** $\min_{s,\pm} E^{s,\pm} \geq 1/\alpha$ **then**

    break ▷ certificate earned early

  end if

end for

run R-VOUCH probes P1 (relearn), P2 (quantise), P3 (jailbreak); repeat the loop above per probe

if $\min_{s,\pm} E^{s,\pm} \geq 1/\alpha$ **and** every probe passes **then**

  issue cert $= (\varepsilon, \alpha, t, C_t, \mathrm{root}, \mathcal{F}, \text{probe report})$

else

  withhold ▷ undetermined: report C_t and continue monitoring

end if

Streaming (waves $k = 1, 2, \ldots$): fresh cohort and coins per wave; global certificate = all-pass; global alarm = $\prod_k R^{(k)}$; per-wave alarms at $\alpha_k = \alpha 2^{-(k+1)}$ ▷ Theorem 6

point: the design does the work, so we give every proof in full. Throughout, $\mathcal{F}_{t-1}$ denotes the verifier's filtration of Section 3.3, generated by everything opened through step $t-1$; all bets and mixture weights are $\mathcal{F}_{t-1}$-measurable (predictable).

A word on what changed in this version, since it is the technical heart of the paper. An earlier draft defined the certified quantity as the *marginal* sign probability $\Pr[Z_i^{(s)} = 1]$ and proved coverage from a supermartingale argument that in fact requires the stronger conditional statement $\mathbb{E}[Z_i^{(s)} \mid \mathcal{F}_{i-1}] \geq p_0$ at every step. The two coincide when signs are independent and identically distributed across pairs, and they come apart when they are not—which is exactly our situation, since every canary passes through one jointly trained and unlearned model. Section 5.1 measures the gap and finds it large. We therefore restate the certificate against the quantity the protocol actually controls, and we obtain a guarantee that needs no independence, no exchangeability across pairs, and no homogeneity across templates or repetition strata.

4.1 The exact null

Theorem 1 (Exact conditional null) *Suppose the unlearned model M_u is conditionally independent of the coin vector $b = (b_1, \ldots, b_m)$ given the pair contents $C = \{(c_i^0, c_i^1)\}$; exact unlearning and retraining-from-scratch both satisfy this. Then for every score $s \in \mathcal{F}$, every step t, and the pair $j = \pi(t)$ opened at that step,*

$$Z_j^{(s)} \mid \mathcal{F}_{t-1} \;\sim\; \mathrm{Bern}(\tfrac{1}{2}) \quad \textit{exactly,}$$

for any model, any score, and any set of query prompts. In particular the sign sequence is a sequence of exact conditional fair coins whether or not the signs are independent across pairs.

Proof Enlarge $\mathcal{F}_{t-1}$ to $\mathcal{G}_{t-1} = \mathcal{F}_{t-1} \vee \sigma\big(\pi(t), \{c_j^0, c_j^1\}\big)$, which reveals the position and the *unordered* pair opened at step t but not its coin; this is legitimate because the manifest is committed leaf-by-leaf (Section 3.3), so b_j is opened only at step t. Given $\mathcal{G}_{t-1}$ the two numbers $s(M_u, x)$ and $s(M_u, y)$ attached to the unordered pair $\{x, y\} = \{c_j^0, c_j^1\}$ are determined. The coin b_j is independent of the other coins and of C by construction, and independent of M_u given C by hypothesis; nothing opened before step t involves b_j. Hence $b_j \mid \mathcal{G}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$. Since b_j decides which of the two determined numbers is the in-twin's,

$$D_j^{(s)} \;=\; s(M_u, c_j^{\mathrm{in}}) - s(M_u, c_j^{\mathrm{ghost}}) \;=\; \pm\big|s(M_u, x) - s(M_u, y)\big|$$

with the two signs equally likely, so $D_j^{(s)} \overset{d}{=} -D_j^{(s)}$ given $\mathcal{G}_{t-1}$. Therefore $\Pr[D_j^{(s)} > 0 \mid \mathcal{G}_{t-1}] = \Pr[D_j^{(s)} < 0 \mid \mathcal{G}_{t-1}]$, and the committed tie-break coin splits any atom at zero equally, giving $Z_j^{(s)} \mid \mathcal{G}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$. Averaging over $\mathcal{G}_{t-1} \supseteq \mathcal{F}_{t-1}$ gives the claim. No property of the model, the score, the query set, or the dependence between pairs was used beyond measurability. □

Theorem 1 is the reason VOUCH needs neither shadow models nor a retrained reference: the null distribution is not estimated or assumed but generated by our own coins, and it is exact at finite m. It is worth being explicit about why the dependence objection does not bite here. The signs $Z_1^{(s)}, \ldots, Z_m^{(s)}$ are indeed dependent—they share one model—but the proof never asks for independence. It asks only that the coin governing the pair opened at step t still be fair given everything opened before, and under exact unlearning it is, because the model carries no information about it. The revocation arm therefore inherits an exact null with no further conditions.

The certificate arm, whose null is the composite "some score still leaks," is where dependence has to be confronted, and that is the next subsection.

4.2 The certified quantity and the certificate

Once M_u and the manifest exist, the sign vector $z^{(s)} = (z_1^{(s)}, \ldots, z_m^{(s)})$ of the committed cohort is fixed; the only randomness left to the audit is the reveal order and the tie coins. We therefore certify the advantage the declared score actually achieves against that cohort.

Definition 1 (Realised cohort advantage) For $s \in \mathcal{F}$ the *realised cohort advantage* is

$$\Delta^{(s)} \;=\; \frac{2}{m}\sum_{i=1}^{m} z_i^{(s)} \;-\; 1 \;=\; 2\ \mathrm{Pr}_I\big[z_I^{(s)} = 1\big] - 1, \tag{10}$$

the advantage of the rule "guess that the twin scoring higher under s is the in-twin" when the pair index I is drawn uniformly from the committed cohort. Two coarser quantities sit above it: the *mechanism-level* advantage $\Delta_C^{(s)} = \mathbb{E}_b[\Delta^{(s)} \mid C, \xi]$, averaging over the inclusion coins with the pair contents C and the training randomness ξ held fixed; and the *super-population* advantage $\bar{\Delta}^{(s)} = \mathbb{E}[\Delta^{(s)}]$, averaging over C and ξ as well.

$\Delta^{(s)}$ is the quantity an auditor of *this* model wants bounded, and it is the one the protocol controls exactly; Definition 2 is stated for it. Proposition 3 climbs one level, from $\Delta^{(s)}$ to $\Delta_C^{(s)}$, at a stated cost in tolerance. We do not climb the second level to $\bar{\Delta}^{(s)}$: that would need concentration over the draw of corpora, which we neither prove nor assume. Where the experiments report a "super-population" verdict they mean the fixed-boundary process of the earlier design, whose null is the predictable conditional mean; we keep both columns throughout so that the reader can see which claims survive at which tolerance.

Definition 2 ((ε, α)-forgetting certificate) Fix a tolerance $\varepsilon \in (0, 1)$, an error budget $\alpha \in (0, 1)$, and the boundary $p_0 = \frac{1}{2} + \frac{\varepsilon}{2}$. An (ε, α)-forgetting certificate for M_u over the class $\mathcal{F}$ is a data-dependent stopping decision that asserts $\sup_{s\in\mathcal{F}} |\Delta^{(s)}| < \varepsilon$, issued by a procedure with the guarantee that, whenever in truth $\sup_{s\in\mathcal{F}} |\Delta^{(s)}| \geq \varepsilon$, the probability that a certificate is *ever* issued—at any stopping time, under any monitoring rule—is at most α.

The two-sidedness in Definition 2 is deliberate: $\Delta^{(s)} \leq -\varepsilon$, the over-forgetting regime in which the in-twin scores conspicuously *below* its never-seen partner, is membership leakage exactly as much as $\Delta^{(s)} \geq \varepsilon$ is, and Section 5 shows that gradient-ascent unlearning lands there in practice.

Theorem 2 (Certificate coverage under arbitrary dependence) *Reveal the cohort in a uniformly random order π drawn by the verifier* independently of the realised sign vectors $\{z^{(s)}\}_{s\in\mathcal{F}}$ *and committed before any query, and for each score s and direction run the wealth* *process of Eq.* (4) *with stakes confined to* $\lambda_t^{s,+} \in [0,\, c/(1-p_0(t)))$ *and* $\lambda_t^{s,-} \in [0,\, c/\bar{p}_0(t))$ *for some fixed* $c < 1$*, against the predictable without-replacement boundary*

$$p_0(t) \;=\; \frac{m\,p_0 \;-\; \sum_{u<t} Z^{(s)}_{\pi(u)}}{m-t+1}, \tag{11}$$

declaring the null refuted outright if ever $p_0(t) > 1$*. Issue the certificate only when* every *per-score, per-direction process exceeds* $1/\alpha$*. Then for any class* $\mathcal{F}$ *and any stopping rule,*

$$\Pr\big[\textit{a false } (\varepsilon,\alpha)\textit{-certificate is ever issued}\big] \le \alpha \qquad \textit{whenever } \sup_{s\in\mathcal{F}} |\Delta^{(s)}| \ge \varepsilon.$$

The guarantee is exact and finite-sample, and it holds conditionally on the cohort's sign vectors*: no independence across pairs, no exchangeability across pairs, and no homogeneity across templates or repetition strata is assumed. No multiplicity correction over* $\mathcal{F}$ *is required. The one thing that* is *assumed is the independence of* π *from the signs, and Remark 1 explains why it is not free.*

Proof Suppose the certification claim is false, so some score s^* and direction violate the tolerance; say $\Delta^{(s^*)} \ge \varepsilon$, i.e. the cohort mean $\mu = \frac{1}{m}\sum_i z_i^{(s^*)}$ satisfies $\mu \ge p_0$ (the case $\Delta^{(s^*)} \le -\varepsilon$ is symmetric, run against $1-p_0$). Condition on the entire sign vector $z^{(s^*)}$; the remaining randomness is the uniform permutation π, and sequential reveal of a uniform permutation is precisely simple random sampling without replacement. Hence the conditional mean of the next sign is the mean of the unrevealed remainder,

$$\mathbb{E}\big[Z^{(s^*)}_{\pi(t)} \,\big|\, \mathcal{F}_{t-1}\big] = \frac{m\mu - \sum_{u<t} Z^{(s^*)}_{\pi(u)}}{m-t+1} \;\ge\; \frac{mp_0 - \sum_{u<t} Z^{(s^*)}_{\pi(u)}}{m-t+1} \;=\; p_0(t),$$

using $\mu \ge p_0$. Each wealth factor therefore has conditional mean

$$\mathbb{E}\big[\,1 + \lambda_t\,(p_0(t) - Z^{(s^*)}_{\pi(t)}) \,\big|\, \mathcal{F}_{t-1}\big] = 1 + \lambda_t\big(p_0(t) - \mathbb{E}[Z^{(s^*)}_{\pi(t)} \mid \mathcal{F}_{t-1}]\big) \;\le\; 1,$$

since $\lambda_t \ge 0$ is predictable. Nonnegativity is the other half and needs the stake restriction: with $Z \in \{0,1\}$ the factor $1+\lambda_t(p_0(t)-Z)$ is minimised at $Z=1$, where it equals $1-\lambda_t(1-p_0(t)) \ge 1-c > 0$ by $\lambda_t < c/(1-p_0(t))$; the mirror arm is identical with $\bar{p}_0(t)$ in place of $1-p_0(t)$. (This bound is why the implementation must derive the stake cap from the *same* clamped boundary the payoff is evaluated at; see Appendix A.) So the process is a nonnegative supermartingale started at one, and Ville's inequality gives $\Pr[\exists t : E_t^{s^*,+} \ge 1/\alpha] \le \alpha$. The refutation branch is never taken under the null, because $p_0(t) \le \mathbb{E}[Z_{\pi(t)} \mid \mathcal{F}_{t-1}] \le 1$. Issuing the certificate requires *every* process to exceed $1/\alpha$—in particular the violating score's own—so a false certificate is issued with conditional probability at most α, uniformly over stopping rules; averaging over the sign vectors preserves the bound. No union bound over $\mathcal{F}$ is needed because the argument runs entirely through the single process attached to the violating score. □

Two consequences are worth naming because they shape how the certificate should be read. First, the guarantee spends no assumption on the *data*: whatever dependence the shared model induces among the signs, it is absorbed by conditioning on the realised sign vector, and the only randomness the proof spends is the verifier's own permutation. Second, and less comfortably, a cohort audited to exhaustion resolves *deterministically*: at $t = m$ the realised mean is known, and no inference remains. The betting machinery therefore buys three things—early stopping at a fraction of the query budget, validity under continuous monitoring and across deletion waves, and the alarm arm, whose null is the genuinely stochastic one of Theorem 1—rather than buying the cohort statement itself. We think this is the honest way to describe what a canary audit can do, and Section 5.4 quantifies each of the three.

Remark 1 (The reveal order must be the verifier's own randomness) Theorem 2 conditions on the sign vector and spends the permutation, so it needs $\pi \perp \{z^{(s)}\}$. That is not a formality here. A provider who can locate the cohort—and Section 5.16 shows that is cheap—could, if it also knew π, arrange which pairs are revealed early. The released implementation therefore draws π from one of three sources, and records which on the certificate. A *public randomness beacon* is the recommended one: the verifier fetches a drand round published *after* the manifest commitment and seeds π with $H(\text{root} \,\|\, \text{beacon randomness})$, so the order is unpredictable to the provider before the round exists, independent of the signs, and reconstructible afterwards by anyone holding the certificate—the audit's own randomness becomes as checkable as its commitment. Failing network access, the verifier draws from operating-system entropy, which satisfies the hypothesis but is not third-party verifiable. The third source is the run seed, which is what produced the results in this paper and which does *not* satisfy the hypothesis, since that seed also seeds cohort generation and training; nothing in the pipeline exploits the coupling, but the condition is not met, and Appendix A records exactly where. Binding the beacon value to the commitment matters in both directions: drawing the beacon first would let a provider grind the manifest against a known order.

Proposition 3 (Transfer to the coin-averaged advantage) *Fix the pair contents C and the training randomness ξ, and write $\Delta_C^{(s)} = \mathbb{E}_b\big[\Delta^{(s)} \,\big|\, C, \xi\big]$ for the advantage averaged over the inclusion coins alone—the advantage the unlearning* mechanism *yields on this corpus. Assume* bounded coin influence*: flipping a single coin b_i changes at most κ of the m signs $z_j^{(s)}$. Then for any $\alpha' \in (0,1)$, with probability at least $1-\alpha'$,*

$$\big|\Delta^{(s)} - \Delta_C^{(s)}\big| \;\le\; \kappa\sqrt{\frac{2\log(2/\alpha')}{m}}. \tag{12}$$

Consequently a cohort certificate at tolerance ε implies a mechanism-level certificate at tolerance $\varepsilon + \kappa\sqrt{2\log(2/\alpha')/m}$, at total error level $\alpha + \alpha'$.

Proof Write $f(b) = \frac{2}{m}\sum_j z_j^{(s)} - 1$ as a function of the coin vector, the other randomness held fixed. Changing one coordinate changes $\sum_j z_j^{(s)}$ by at most κ by hypothesis, so f has bounded differences $c_i = 2\kappa/m$. McDiarmid's inequality gives $\Pr[|f - \mathbb{E}_b f| \ge u] \le 2\exp(-2u^2/\sum_i c_i^2) = 2\exp(-u^2 m/(2\kappa^2))$, with the expectation over the coins and (C,ξ) held fixed—which is exactly $\Delta_C^{(s)}$; setting the right-hand side to α' yields (12). □

Three qualifications travel with Proposition 3 and we would rather state them than let a reader over-read the bound. First, it bounds the distance to $\Delta_C^{(s)}$, the *coin-averaged* advantage on this corpus, not to $\bar{\Delta}^{(s)} = \mathbb{E}[\Delta^{(s)}]$ of Definition 1, which additionally averages over the draw of C and ξ; closing that second gap requires a further concentration step over corpora that we do not have and do not claim. Second, the bound is informative only when $\kappa = o(\sqrt{m})$: at $\kappa = \Theta(m)$ it is vacuous. Under exact unlearning $\kappa = 1$ holds trivially, since a coin that leaves no trace in the model can only affect its own pair's sign; away from it, an influential coin can move the model and hence many signs, which is precisely the regime where the transfer is wanted. Third, the size of the price is worth internalising: at $m = 512$, $\alpha' = 0.05$ and $\kappa = 1$ the inflation is 0.12, which is why a mechanism-level claim at $\varepsilon = 0.05$ needs a cohort an order of magnitude larger than a cohort claim at the same tolerance—a gap Section 5.8 reports rather than elides.

Remark 2 (Why the intuitive mixture fails) It is tempting to run a *single* certificate process on the adaptively reweighted sign $\bar{Z}_t = \sum_s w_s Z_t^{(s)}$, and indeed this is valid for the revocation arm, whose null makes every component an exact conditional fair coin. It is *not* valid for the certificate, whose composite null is "some score still leaks." The failure is easy to exhibit: take $\Delta^{(s_1)} = \varepsilon$ exactly at the boundary and $\Delta^{(s_2)} = \cdots = 0$ clean. Then $\mathbb{E}[\bar{Z}_t \mid \mathcal{F}_{t-1}] = w_1 p_0 + (1 - w_1)/2 < p_0$ whenever $w_1 < 1$, so each factor of the mixture's certificate process has conditional mean strictly greater than one: the process is a *sub*martingale under the null and crosses $1/\alpha$ with probability approaching one. Worse, the violation is generic rather than adversarial, because discrimination-driven reweighting moves weight *away* from the one leaking score under partial unlearning. Against this construction—which is our own earlier design, not a published method—we measured a false-certification rate of 0.996 against 0.007 for the per-score minimum (Table 4), which is why VOUCH requires consensus rather than an average.

4.3 Power, confidence sequences, and composition

Theorem 4 (Power, fixed-boundary process) *Consider the* super-population *arm, i.e. the process of Eq. (4) run against the fixed boundary $p_0(t) \equiv p_0$ with signs i.i.d. across pairs. If $|\Delta^{(s)}| < \varepsilon$ for all s and the bets have sublinear regret (the mixture of Section 3.5 qualifies), then $\frac{1}{t} \log E_t \to \mathrm{KL}(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(p_0))$ almost surely for each process, and the expected certification time is $\mathbb{E}[\tau^*] = (1+o(1)) \log(1/\alpha)/\min_{s,\pm} \mathrm{KL}$, the expectation being taken under exact unlearning and the asymptotics in small ε. Since*

$$\mathrm{KL}\big(\mathrm{Bern}(\tfrac{1}{2}) \,\big\|\, \mathrm{Bern}(\tfrac{1}{2} + \tfrac{\varepsilon}{2})\big) \;=\; -\tfrac{1}{2} \log\big(1 - \varepsilon^2\big) \;=\; \tfrac{\varepsilon^2}{2} + O(\varepsilon^4), \tag{13}$$

this is approximately $2 \log(1/\alpha)/\varepsilon^2$ pairs.

The restriction to the fixed boundary is not cosmetic. Under without-replacement reveal the log-wealth increments are neither independent nor identically distributed, the boundary $p_0(t)$ moves, and the admissible stake range moves with it—so "the best constant bet" is not well defined, and neither the $O(\log t)$ ONS regret bound (which assumes a fixed convex domain) nor the Krichevsky–Trofimov redundancy bound (which assumes a fixed p_0) transfers as stated. Theorem 4 should therefore be read as sizing the cohort for a *mechanism-level* claim. For the finite-cohort target we do not have a matching analysis; its certification time is capped by m, since the claim resolves deterministically once the cohort is exhausted, and we report its behaviour empirically instead (Table 6, cohort column).

Proof Fix one process, say the upper process for a score with true sign probability $p < p_0$. For a constant bet λ, the log-wealth increments $\log(1 + \lambda(p_0 - Z_t))$ are i.i.d. with mean

$f(\lambda) = p\log(1-\lambda(1-p_0)) + (1-p)\log(1+\lambda p_0)$, so by the strong law $\frac{1}{t}\log E_t \to f(\lambda)$ almost surely. The Kelly-optimal constant bet is $\lambda^* = (p_0 - p)/(p_0(1-p_0))$: substituting it turns the two possible wealth factors into the Bernoulli likelihood ratios $1-\lambda^*(1-p_0) = p/p_0$ and $1+\lambda^* p_0 = (1-p)/(1-p_0)$, so $f(\lambda^*) = p\log\frac{p}{p_0} + (1-p)\log\frac{1-p}{1-p_0} = \mathrm{KL}(\mathrm{Bern}(p)\,\|\,\mathrm{Bern}(p_0))$, and no constant bet can do better because f is concave with this maximum. A betting strategy whose log-wealth regret against the best constant bet is $o(t)$ therefore attains the same limit; the ONS expert has $O(\log t)$ regret, the KT plug-in (6) achieves $\mathrm{KL} - \frac{1}{2t}\log t$ directly by the Krichevsky–Trofimov redundancy bound, and the uniform mixture over eight experts trails its best expert by at most the constant $\log 8$, so all three satisfy the hypothesis. For the crossing time, the process rejects when $\log E_t \geq \log(1/\alpha)$; since $\frac{1}{t}\log E_t \to \mathrm{KL}$ almost surely, the first-crossing time τ^* satisfies $\mathbb{E}[\tau^*] = (1+o(1))\log(1/\alpha)/\mathrm{KL}$ by the standard Wald argument, and issuance waits for the *slowest* process, whose drift is $\min_{s,\pm}\mathrm{KL}$. Finally, under exact unlearning $p = \frac{1}{2}$ for every score, and

$$\mathrm{KL}\big(\mathrm{Bern}(\tfrac{1}{2})\,\|\,\mathrm{Bern}(p_0)\big) = \tfrac{1}{2}\log\frac{1/2}{p_0} + \tfrac{1}{2}\log\frac{1/2}{1-p_0} = -\tfrac{1}{2}\log\big(4p_0(1-p_0)\big) = -\tfrac{1}{2}\log(1-\varepsilon^2),$$

whose expansion is (13), giving $\mathbb{E}[\tau^*] \approx 2\log(1/\alpha)/\varepsilon^2$. □

Theorem 4 tells the provider how large a canary cohort to plant. An $\varepsilon = 0.1$ certificate at $\alpha = 0.05$ needs on the order of 600 pairs, an $\varepsilon = 0.05$ certificate roughly 2,400, and an $\varepsilon = 0.02$ certificate roughly 15,000; because the drift equals the Kullback–Leibler divergence—the information-theoretic optimum for testing $\mathrm{Bern}(\frac{1}{2})$ against $\mathrm{Bern}(p_0)$—no valid procedure can do better by more than a constant factor. We note that an earlier version of this theorem reported the expansion as $\varepsilon^2/8$ and the sample size as $8\log(1/\alpha)/\varepsilon^2$; (13) corrects it. The numerical columns of Table 6 were computed from the exact divergence and are unaffected.

Theorem 5 (Finite-cohort certification time) *Fix a score s and run the upper certificate process of Theorem 2: the without-replacement boundary (11) with the rule that the null is refuted outright once $p_0(t) > 1$. Let the committed cohort of size m carry exactly j signs $z_i^{(s)} = 1$, and let $k_0 = \lceil m p_0 \rceil$ be the least number of such signs compatible with the null $\Delta^{(s)} \geq \varepsilon$. If $j \leq k_0 - 2$—the realised cohort sits at least two pairs below the boundary—then the process refutes, and hence certifies at* any *α, by the stopping time*

$$\tau^* \;\leq\; m - (k_0 - j) + 2 \qquad \textit{for every reveal order } \pi, \tag{14}$$

and, under the uniformly random π that Theorem 2 requires,

$$\mathbb{E}[\tau^*] \;\leq\; \frac{(m-k_0+1)(m+1)}{m-j+1} \;+\; 1, \tag{15}$$

with equality when the decision is reached by refutation. The lower arm is the mirror statement with z replaced by $1-z$ and p_0 by $1-p_0$; the two-sided certificate over a class $\mathcal{F}$ is issued by the largest of the $2|\mathcal{F}|$ individual bounds. Neither bound involves α, the betting strategy, or $|\mathcal{F}|$.

Proof Refutation fires at the first t with $p_0(t) > 1$, i.e. with $mp_0 - \sum_{u<t} Z_{\pi(u)} > m - t + 1$. Writing S_{t-1} for the revealed count of ones and $W_{t-1} = (t-1) - S_{t-1}$ for the revealed count of zeros, and using $S_{t-1} + (m-t+1) = m - W_{t-1}$, the condition is $m - W_{t-1} < mp_0$, that is $W_{t-1} > m(1-p_0)$. Both sides being compared to an integer, this is exactly $W_{t-1} \geq$

$m - k_0 + 1 =: r$, whether or not mp_0 is an integer. Refutation therefore occurs one step after the r-th zero is revealed: $\tau^* = 1 + \pi$-position of the r-th zero. The cohort contains $m - j$ zeros, and $r \le m - j - 1$ exactly when $j \le k_0 - 2$, so under that hypothesis the r-th zero is revealed at some position $\le m - 1$ and $\tau^* \le m$.

For (14), the position of the r-th zero is largest when every one precedes every zero, putting it at $j + r = j + m - k_0 + 1$; adding the one step gives the bound, and the ordering that attains it shows it is tight. For (15), let $M = m - j$ be the number of zeros. In a uniformly random arrangement the M zeros split the $m - M$ ones into $M + 1$ exchangeable gaps, so the expected number of ones preceding the r-th zero is $r(m - M)/(M + 1)$ and the expected position of that zero is $r + r(m - M)/(M + 1) = r(m + 1)/(M + 1)$. Adding one step gives (15).

Finally, the argument never inspects the stake λ_t: refutation is a statement about which cohort compositions remain feasible, so it holds for every predictable betting strategy, and the certificate's actual stopping time is the minimum of this time and the time its wealth crosses $1/\alpha$ by ordinary growth. □

Theorem 5 is the finite-population counterpart that Theorem 4 could not supply. Under without-replacement reveal the log-wealth increments are neither independent nor identically distributed and the regret bounds behind Theorem 4 do not transfer, so we bound the cohort target's stopping time by a different route: not by how fast wealth grows, but by when the null's remaining feasible compositions run out. The two mechanisms run in parallel and the process stops at whichever fires first, which is what makes the cohort column of Table 6 legitimately faster than the Kullback–Leibler column rather than mysteriously so. Their regimes are also easy to tell apart: exponential growth dominates when the cohort is large relative to $\log(1/\alpha)/\mathrm{KL}$, and the feasibility bound binds when it is not, which is exactly the tight-tolerance regime of Section 5.11. At $m = 384$, $\varepsilon = 0.2$ and a cohort at chance ($j = m/2$), (15) gives 308 pairs against a measured median of 165: the wealth process usually wins, and the feasibility bound is the guarantee that holds when it does not.

Remark 3 (Per-score versus simultaneous confidence sequences) Each score's confidence sequence (8) covers that score's own advantage $\Delta^{(s)}$ with uniform-in-time probability $1 - \alpha$; the guarantee is *per score*. Two derived statements should be distinguished. The reported $\max_{s \in \mathcal{F}} U_t^{(s)}$ *is* a valid level-α anytime upper bound on $\sup_{s \in \mathcal{F}} \Delta^{(s)}$ with no multiplicity correction, because the maximising index of the population quantities is a fixed, non-random index once the cohort is fixed, and that single score's own sequence is the only one the argument needs. Simultaneous two-sided coverage of *all* $|\mathcal{F}|$ advantages, on the other hand, costs a union bound and holds at level $|\mathcal{F}|\alpha$; we report intervals at that corrected level wherever we display the whole class.

Theorem 6 (Streaming composition, in both directions) *Let deletion waves $k = 1, \ldots, K$ use cohorts with independent coins.* (a) Certificates compose by consensus. *Claiming that the whole history is clean only when every wave has individually earned its certificate keeps the probability of a false history-wide claim at or below α for any K; multiplying certificate e-values instead tests the intersection null "every wave is bad," whose rejection certifies merely that* some *wave is clean.* (b) Alarms compose by product. *Under the intersection null "every wave was exactly unlearned" each per-wave revocation process is a supermartingale, so their product is one too and crosses $1/\alpha$ with probability at most α; the product accumulates leakage* *spread so thinly across waves that no single wave reveals it.* (c) *Per-wave alarms with α-spending $\alpha_k = \alpha 2^{-(k+1)}$ control family-wise false revocation over an unbounded history.*

Proof *(a)* Suppose the history-wide claim is false, so some wave k^* truly has $\sup_s |\Delta^{(s),(k^*)}| \geq \varepsilon$. The all-pass rule claims only when every wave has individually issued, so claiming requires wave k^*'s own per-score minimum process to cross $1/\alpha$, an event of probability at most α by Theorem 2—independently of K and of how the auditor schedules the waves. For the negative half of the statement, note that a product $\prod_k E^{(k)}$ of *certificate* e-values is, by construction, an e-value for the intersection null $\bigcap_k \{\text{wave } k \text{ bad}\}$; rejecting an intersection establishes only the disjunction "some wave is clean." In the one-bad-wave scenario the clean waves' e-values grow without bound and drag the product over $1/\alpha$ regardless of the bad wave. We measure the damage rather than assert it: over 500 ten-wave histories containing one genuinely bad wave, the product rule issues a false history-wide certificate in *every* history whose bad wave truly violates the tolerance, against 0.000 for the all-pass rule (Table 7).

(b) Under the intersection null, Theorem 1 applies wave by wave, so each $R^{(k)}$ is a nonnegative supermartingale with initial value one with respect to the combined filtration, and cohorts across waves are independent by construction. Conditional on $\mathcal{F}_{t-1}$, the increment of the product $\prod_k R^{(k)}$ at any arrival is the increment of the single wave whose pair is revealed, which has conditional mean at most one; the product is therefore itself a nonnegative supermartingale started at one, for any adapted within-wave sampling times, and Ville's inequality bounds its crossing probability by α. Its power against distributed leakage is additive in the exponent: if every wave carries advantage $\Delta_0 > 0$, each factor drifts upward at rate $\mathrm{KL}(\mathrm{Bern}(\frac{1}{2} + \frac{\Delta_0}{2}) \,\|\, \mathrm{Bern}(\frac{1}{2}))$ per pair, so the product crosses after roughly $\log(1/\alpha)/(m \cdot \mathrm{KL})$ waves even when no single wave is individually detectable.

(c) Give wave k its own alarm at level $\alpha_k = \alpha 2^{-(k+1)}$. By Ville applied per wave, the probability that a truly clean wave k ever alarms is at most α_k, and the union bound over an unbounded horizon costs $\sum_{k \geq 1} \alpha 2^{-(k+1)} \leq \alpha/2 \leq \alpha$. □

Proposition 7 (A magnitude-aware alarm) *Under the null of Theorem 1,* $\mathrm{sign}(D_t)$ *is a fair coin even conditional on* $|D_t|$*, so wealth factors of the form* $1 + \lambda_t \,\mathrm{sign}(D_t)\, g_t$ *are an exact e-process for any predictable stake* $g_t \in [0, 1]$ *measurable with respect to the past and* $|D_t|$*. Taking* g_t *to be the predictable rank of* $|D_t|$ *concentrates bets on the large-effect pairs where sparse memorisation lives, and halves detection time under* 5%*-sparse leakage (Section 5.18). Because it bets aggressively we report it as an optional power enhancement and keep the sign-based arm as the default.*

Proof Symmetry of the conditional law of D_t (Theorem 1) means that conditioning on the magnitude $|D_t|$ leaves the sign exactly fair: $\mathbb{E}[\mathrm{sign}(D_t) \mid |D_t|, \mathcal{F}_{t-1}] = 0$, with ties randomised by the committed coin. For any stake $g_t \in [0, 1]$ that is measurable with respect to $\mathcal{F}_{t-1} \vee \sigma(|D_t|)$—the predictable rank of $|D_t|$ among past magnitudes qualifies—the tower property gives $\mathbb{E}[\mathrm{sign}(D_t)\, g_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\, g_t \,\mathbb{E}[\mathrm{sign}(D_t) \mid |D_t|, \mathcal{F}_{t-1}] \mid \mathcal{F}_{t-1}] = 0$, so each factor $1 + \lambda_t \,\mathrm{sign}(D_t)\, g_t$ with predictable $\lambda_t \in [0, 1)$ has conditional mean exactly one and stays positive. The product is a nonnegative martingale with initial value one, hence an exact e-process, and Ville's inequality applies as before. The power claim is empirical and quantified in Section 5.18. □

Proposition 8 (One-directional bridge from certified unlearning) *If $\mathcal{U}$ is $(\varepsilon_u, \delta_u)$-certified, then $|\bar{\Delta}^{(s)}| \le \frac{e^{\varepsilon_u}-1}{e^{\varepsilon_u}+1} + \delta_u$ for every s, so for ε above this value VOUCH issues with probability $1 - \alpha - o(1)$. A certified algorithm therefore passes VOUCH, as consistency demands, while VOUCH remains meaningful at scales where $(\varepsilon_u, \delta_u)$ certificates are unattainable.*

Proof The sign $Z^{(s)}$ is a randomised post-processing of the pipeline output: it is computed from M_u and the pair alone. $(\varepsilon_u, \delta_u)$-certified unlearning makes the law of M_u, under either value of the coin, $(\varepsilon_u, \delta_u)$-indistinguishable from the law of the retrained reference, under which $Z^{(s)}$ is an exact fair coin by Theorem 1. Now view the rule "guess that the twin with the larger score is the in-twin" as a hypothesis test for the coin b_i; its advantage is exactly Pr[correct] − Pr[wrong] $= \bar{\Delta}^{(s)}$. The trade-off region of Kairouz et al. [49] states that any test between two $(\varepsilon_u, \delta_u)$-indistinguishable distributions has type-I and type-II errors satisfying $e^{\varepsilon_u}\beta_1 + \beta_2 \ge 1 - \delta_u$ and $\beta_1 + e^{\varepsilon_u}\beta_2 \ge 1 - \delta_u$; adding the two constraints, $\beta_1 + \beta_2 \ge 2(1 - \delta_u)/(1 + e^{\varepsilon_u})$, so the advantage obeys

$$|\bar{\Delta}^{(s)}| \;\le\; 1 - (\beta_1 + \beta_2) \;\le\; 1 - \frac{2(1-\delta_u)}{1 + e^{\varepsilon_u}} \;=\; \frac{e^{\varepsilon_u} - 1}{e^{\varepsilon_u} + 1} + \frac{2\delta_u}{1 + e^{\varepsilon_u}} \;\le\; \frac{e^{\varepsilon_u} - 1}{e^{\varepsilon_u} + 1} + \delta_u.$$

When ε strictly exceeds this bound, every certificate process has strictly positive drift by Theorem 4 and crosses $1/\alpha$ with probability tending to one as $m \to \infty$, while the revocation arm false-alarms with probability at most α; the issuance probability is therefore at least $1 - \alpha - o(1)$. □

Remark 4 (The bridge runs one way only) Proposition 8 states that certified unlearning implies passing VOUCH. The converse is false and we claim nothing like it: passing VOUCH does *not* imply any $(\varepsilon_u, \delta_u)$ guarantee, does not bound the influence of any datum outside the canary cohort, and does not constrain scores outside $\mathcal{F}$. The proposition is a consistency check on the definition—an algorithm with a parameter-space guarantee should not be flagged by a behavioural audit—and it should not be read as a semantic equivalence in either direction.

Scope, stated precisely.

A VOUCH certificate concerns the advantage extractable through the declared class $\mathcal{F}$, on the committed canary cohort, against the deployed model, under the honest-but-verifiable threat model of Section 3.1. It does not assert erasure in parameter space, which no behavioural method can; it is a per-cohort statement, not a per-deletion-request one, because per-request certificates would require per-user canaries; the transfer to the super-population advantage costs the explicit inflation of Proposition 3; and the extrapolation from planted canaries to organic records is *calibrated* through the dose–response strata and the head-to-head comparisons of Section 5.12 rather than proven. We regard stating these limits precisely as part of the contribution, since a certificate whose scope is overclaimed is worse than none.

5 Experiments

We organise the study into three tiers so that each claim is tested where it can be tested most sharply. The *simulation tier* draws pair signs directly from their generating distribution—exactly the object Theorems 1 and 2 control—which lets us calibrate validity and power over thousands of seeds without confounding them with the vagaries of a particular model. The *end-to-end tier* fine-tunes real models on corpora with injected canaries, unlearns them with each subject method, and certifies the result, so that the abstract null meets an actual training pipeline. The *benchmark tier* repeats the end-to-end protocol on the field's standard testbeds, TOFU and MUSE-News, across three architectures.

Unless noted we use $\alpha = 0.05$, two-sided certification, the sign-based revocation arm, the finite-cohort certificate target, LoRA fine-tuning of rank 16, $Q = 2$ query wrappers, and between 384 and 640 canary pairs; the full hyperparameter table is Appendix A, and the headline tolerance is $\varepsilon = 0.2$ with the complete sweep over $\varepsilon \in \{0.05, 0.1, 0.2\}$ reported in Table 11. The entire study ran on free, ephemeral compute—Colab and Kaggle GPU sessions, Colab CPU sessions, an Apple-silicon laptop, and a four-core container—coordinated by a released fault-tolerant scheduler; because every training stage checkpoints and every verification is saved, a reclaimed session costs at most 100 optimiser steps.

5.1 Which null is actually controlled

We begin with the question that reorganised this version of the paper. Signs from all canaries pass through one jointly trained and unlearned model, so they are neither independent nor identically distributed across templates and repetition strata. Does that matter? Table 1 answers it directly. We generate sign streams whose *marginal* sign probability is exactly $p_0 = 0.55$ under four regimes and ask how often each certificate process issues.

The answer is that it matters, and in the direction least convenient for the earlier analysis. When signs are independent, the fixed-boundary process of the earlier draft holds its level at 0.034, as its stated theorem claims. Heterogeneity across strata alone does not break it either (0.026), because shuffling the reveal order leaves the realised cohort mean tightly concentrated. But once the model itself carries a shared random effect—so that a given training run either leaks a lot or leaks little—the realised cohort advantage spreads out (its standard deviation rises from 0.031 to 0.50) and the fixed-boundary process issues under the marginal null in 38.9% and 48.1% of runs, eight- and ten-fold breaches of $\alpha = 0.05$. The without-replacement process, by contrast, controls the realised-cohort error at or below 0.020 in every regime, exactly as Theorem 2 requires.

Two readings of that table are available and we want to be clear which one we take. One could say the marginal null is the target and the procedure fails; we say instead that the marginal null is the wrong target, because it averages over training runs the auditor is not auditing, and a certificate should concern the deployed model. The column marked † makes the distinction concrete: those issuances are not errors under the cohort claim, because on those runs the realised advantage genuinely sits below the tolerance. What we do concede without qualification is that the earlier draft claimed the marginal statement while proving the conditional one, and that the gap is not academic.

Table 1 Which null is controlled. Sign streams with *marginal* mean $p_0 = 0.55$ ($\varepsilon = 0.1$), 1,024 pairs, 2,000 seeds, peeking after every pair. "marginal" is the issuance rate under the marginal null; "cohort" restricts to runs whose realised cohort advantage genuinely violates the tolerance, the event Theorem 2 controls. Bold marks a breach of $\alpha = 0.05$. †These issuances are not errors under the cohort claim.

		fixed boundary (v1)		without replacement (ours)	
sign-generating regime	sd($\hat{\Delta}$)	marginal	cohort	marginal	cohort
independent	0.031	0.034	0.012	$0.501^{\dagger}$	0.014
heterogeneous by stratum	0.030	0.026	0.016	$0.495^{\dagger}$	0.020
model-level over-dispersion	0.296	**0.389**	0.001	$0.483^{\dagger}$	0.001
shared model-level latent	0.500	**0.481**	0.000	$0.481^{\dagger}$	0.000

Table 2 Direct verification of Theorem 2: realised type-I error on fixed sign vectors at the least favourable feasible point of the cohort null, revealed in a uniformly random order with peeking after every pair (2,000 seeds, $\alpha = 0.05$). "ours" is the without-replacement process, "fixed" the fixed-boundary one.

		$m = 384$		$m = 1024$	
ε	cohort structure	ours	fixed	ours	fixed
0.1	contiguous	0.028	0.018	0.024	0.014
	spread	0.025	0.015	0.033	0.019
	stratified (rates 0.2–0.9)	0.024	0.015	0.035	0.018
0.2	contiguous	0.026	0.013	0.026	0.013
	spread	0.026	0.010	0.034	0.018
	stratified (rates 0.2–0.9)	0.025	0.013	0.032	0.015

Table 2 verifies the corrected theorem directly, on fixed sign vectors sitting at the least favourable feasible point of the cohort null and carrying deliberately adversarial internal structure—all successes contiguous, spread evenly, or split into four strata with rates from 0.2 to 0.9. Under peeking after every pair the realised error never exceeds 0.035 against a nominal 0.05, for every structure, both tolerances and both cohort sizes. This is the guarantee that needs no assumption at all.

Is the dependence real on real models? Yes. Across the benchmark runs a chi-square test of sign-rate homogeneity across repetition strata rejects at the 5% level in 14% of all runs—and in 24 of the 32 un-unlearned runs, where there is leakage to be heterogeneous about. Heterogeneity across canary templates rejects in 8%. The regime of Table 1 is therefore the empirically relevant one.

5.2 Validity under adversarial optional stopping

With the target settled we can ask whether VOUCH holds its error level when an adversary peeks after every observation and stops at the first crossing. We simulate

Table 3 Type-I error over 2,000 seeds with peeking after every pair (1,024 pairs). Bold marks a procedure whose realised error exceeds its nominal level.

procedure	$\alpha = 0.05$	$\alpha = 0.01$
VOUCH certificate (boundary null $\Delta = \varepsilon$)	0.031	0.008
VOUCH revocation (exact unlearning)	0.030	0.006
binomial w/ peeking — false certification	**0.354**	**0.121**
binomial w/ peeking — false alarm	**0.368**	**0.117**

exactly this attack. Table 3 reports the realised type-I error over 2,000 seeds: VOUCH stays below the nominal level for both the certificate and the revocation arm at both $\alpha = 0.05$ and $\alpha = 0.01$, whereas a fixed-sample binomial test used in the same peeking fashion inflates its error to 0.35, 7 times the nominal rate. The calibration curve in Figure 2 confirms that the conservatism holds across the whole range of α. When we run the complete protocol—three correlated scores, the confidence sequence, and both arms together—the false revocation rate is 0.025 and the per-score confidence sequence covers the truth with uniform-in-time probability 0.971 against the nominal 0.95; and when there is genuine memorisation to find ($\Delta \approx 0.36$), the revocation arm fires in every run at a median of 44 pairs, with no false certificate ever issued.

One caveat on reading these numbers, which a reviewer rightly pressed. A realised error of 0.031 against a nominal 0.05 is what a supermartingale bound is *supposed* to produce; it is evidence of validity, not of tightness, and the same conservatism costs power, which is the price Table 5 quantifies. We do not claim the bound is tight and the gap is not a defect.

5.3 Sound composition across scores

We next confirm the composition result of Theorem 2 and the failure it warns against. Constructing a composite null in which one of three scores sits exactly at the boundary while the other two are clean, we compare the per-score-minimum certificate that VOUCH uses against the adaptive-mixture certificate that seems natural. The difference is reported in Table 4: the mixture arm certifies the leaking model in 99.6% of runs—it is not merely loose but anytime-invalid, as Remark 2 explains—while the per-score minimum holds at 0.7%, comfortably below α. We want to be precise about what this number is and is not: the mixture arm is *our own* earlier design, evaluated against an adversarially constructed composite null. It is not a published method and the figure is not a refutation of anyone else's work.

5.4 Comparison against valid sequential procedures

A fixed-sample binomial test inspected after every pair is a useful illustration of why naive peeking is invalid, but it is not a serious competitor. The right question is what anytime validity costs relative to procedures that are *also* valid, so we add four: exact group-sequential α-spending with Pocock and O'Brien–Fleming boundaries, calibrated by dynamic programming over the exact binomial lattice rather than by a normal

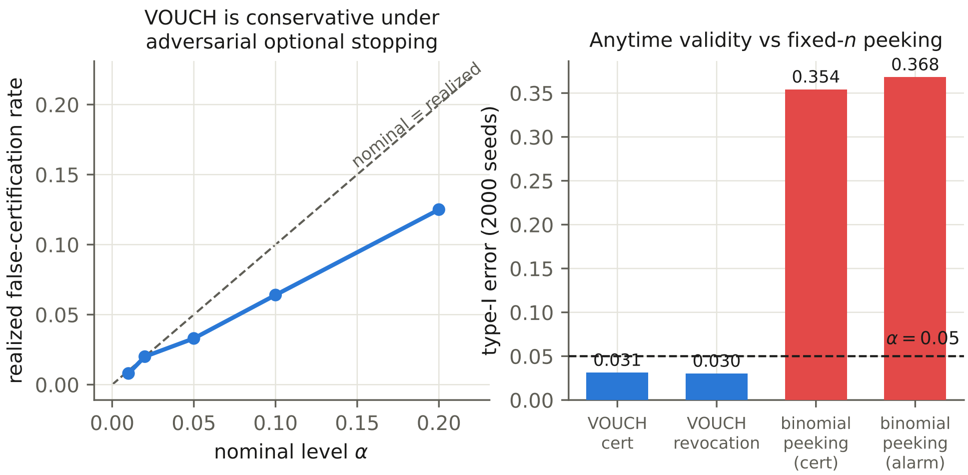

Fig. 2 **Anytime validity.** Left: the realised false-certification rate lies below the nominal level at every α under adversarial optional stopping. Right: a fixed-n binomial test used with peeking inflates its error sevenfold; VOUCH does not.

Table 4 False certification under a composite null (score A at the boundary $p_0 = 0.55$, scores B and C clean; 2,048 pairs, $\alpha = 0.05$). The mixture arm is our own earlier design, not a published method.

certificate arm	false-certification rate
per-score minimum (ours)	0.007
adaptive-mixture sign (design v1.0)	**0.996**

approximation; a truncated Beta$(\frac{1}{2}, \frac{1}{2})$-mixture (Jeffreys) e-process; a Robbins normal-mixture e-process; and a fixed-n binomial test at a pre-committed sample size. All see the same sign stream and the same query budget. Table 5 reports the result.

Read the two type-I columns together, because they measure different things and the distinction is the same one Section 5.1 draws. Against the *marginal* null every super-population procedure holds its level to within Monte Carlo error at 2,000 seeds (0.030 to 0.053, standard error ≈ 0.005), and VOUCH's cohort arm does not—it issues in half of runs, because half of those cohorts have a realised advantage genuinely below the tolerance. Against the *cohort* null every procedure holds its level, and the two group-sequential boundaries and the fixed-n test are markedly conservative there (0.004, 0.000, 0.000) precisely because they were built for the other target. Neither column alone is a fair scoreboard, which is why we print both. On power and cost the ordering is informative. Against exact unlearning at $\varepsilon = 0.1$ and a budget of 1,536 pairs, VOUCH on the cohort target certifies every run using a mean of 523 queries; Pocock reaches 0.971 using 645; O'Brien–Fleming 0.985 using 844; the two mixture e-processes trail at 0.77–0.78; and the fixed-n test attains 0.989 but spends

Table 5 Valid sequential comparators on the same sign stream and query budget (2,000 seeds, $\alpha = 0.05$). "type I" is measured at the least favourable null; for the cohort target it is restricted to runs whose realised cohort advantage violates the tolerance. "power" and "queries" are measured under exact unlearning.

	type I at the least favourable null		under exact unlearning		
procedure	marginal	cohort	power	queries	median t
VOUCH, cohort target	**0.495**	0.018	1.000	523	497
VOUCH, super-population target	0.030	0.016	0.882	747	612
Beta($\frac{1}{2}, \frac{1}{2}$)-mixture e-process	0.040	0.028	0.769	900	715
normal-mixture e-process	0.034	0.037	0.782	901	704
group sequential, Pocock	0.050	0.004	0.971	645	614
group sequential, O'Brien–Fleming	**0.051**	0.000	0.985	844	768
fixed-n binomial at pre-committed n	**0.053**	0.000	0.989	1536	1536

the entire budget of 1,536 queries on every run, because a fixed-n test cannot stop early by construction. The honest summary is that anytime validity is close to free here: against the strongest valid sequential comparator VOUCH gives up nothing in power and saves about a fifth of the queries, and it matches the fixed-n test's power to within a rounding of one point (0.9995 against 0.9885) at a threefold reduction in verifier cost. The comparison also isolates how much of VOUCH's advantage comes from retargeting: on the super-population target, at the same budget, power falls from 1.000 to 0.882 and mean queries rise from 523 to 747. That gap is the price of the stronger claim, and it is the same gap Proposition 3 charges in tolerance.

5.5 Power, tightness, and the cost of a horizon

Table 6 reports certification time against the Kullback–Leibler limit. Two corrections to the earlier version are folded in. First, the theoretical column now follows the corrected expansion (13); the numbers themselves are unchanged because they were always computed from the exact divergence, but the accompanying sample-size rule of thumb is $2 \log(1/\alpha)/\varepsilon^2$, not $8 \log(1/\alpha)/\varepsilon^2$. Second, and more consequentially, the earlier table's starred entries were medians taken over the runs that happened to issue within a 20,000-pair horizon, which is a selection-biased statistic: at $\varepsilon = 0.02$ fully 45% of runs are censored, and conditioning on the fast half produced a median of 11,189 that sat *below* the information-theoretic limit of 14,976, an impossibility that should have been a red flag. Reporting the censoring rate and the Kaplan–Meier median removes the artefact: the corrected median is 18,377, comfortably above the limit, and at $\alpha = 0.01$ censoring reaches 72% and the survival curve never reaches one half, so no median exists and we report none.

The finite-cohort column is faster than the super-population one at every tolerance, and the reason is structural rather than statistical: the cohort claim resolves deterministically once the cohort is exhausted, so its sample complexity is capped by

Table 6 Median pairs to certification under exact unlearning, $\alpha = 0.05$, horizon 20,000 pairs, 500 seeds per cell. For the super-population target the median over issued runs is selection-biased where censoring is heavy, so the Kaplan–Meier median and the censoring rate are given. The KL column is the information limit for the *super-population* target only: the cohort claim resolves once the cohort is exhausted, so its time is capped by m and may legitimately fall below that limit. At $\alpha = 0.01$ (not shown) censoring at $\varepsilon = 0.02$ reaches 72% and the survival curve never reaches one half, so no median exists.

	super-population target			cohort target	
ε	median|issued	Kaplan–Meier	censored	median	$\log(1/\alpha)/\mathrm{KL}$
0.02	11189	18377	**0.45**	10917	14976
0.05	2998	2998	0.00	2690	2394
0.1	686	685	0.00	670	596
0.2	166	165	0.00	165	147

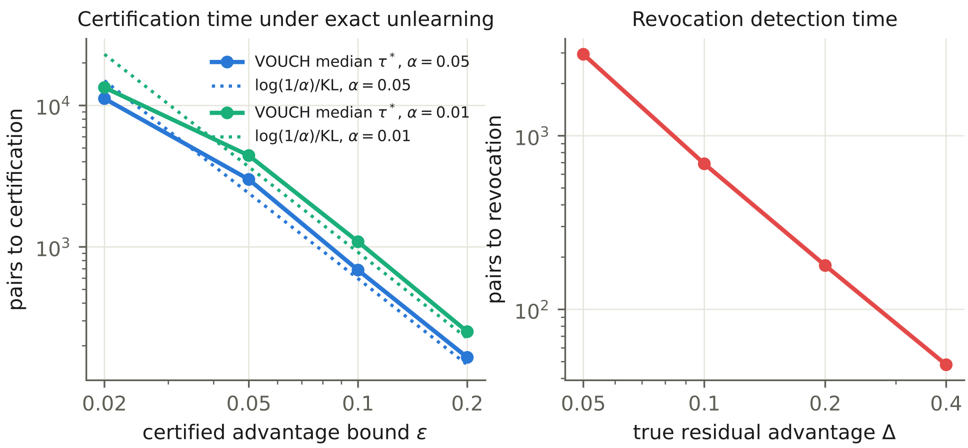

Fig. 3 **Power.** Left: median certification time versus ε, against the information limit $\log(1/\alpha)/\mathrm{KL}$. Right: revocation detection time versus the true residual advantage.

m and it may legitimately sit below a bound that governs the super-population parameter. Readers should not read the cohort column as beating an information limit; it is answering an easier question, and Proposition 3 prices the difference.

Two-sided certification, which we adopt because of the over-forgetting phenomenon documented below, costs roughly a factor of 1.7 in pairs—283 against 166 at $\varepsilon = 0.2$—a price we consider well worth paying for a certificate that cannot be defeated by driving scores below chance. On the detection side the revocation arm finds an advantage of $\Delta = 0.4$ in a median of 48 pairs and $\Delta = 0.1$ in 690. The confidence sequence tightens at the expected $O(\sqrt{\log\log t/t})$ rate (Figure 4): under exact unlearning its upper bound on Δ falls from 0.26 at 128 pairs to 0.10 at 1,000.

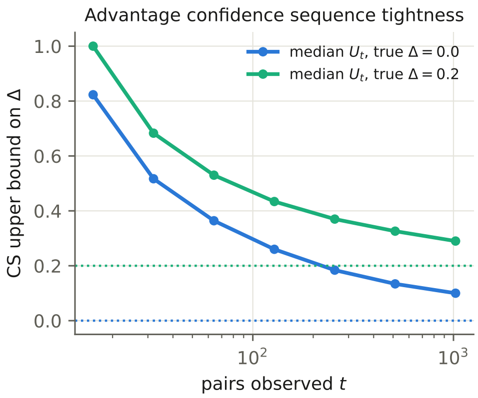

Fig. 4 **Tightness.** The anytime confidence-sequence upper bound on Δ shrinks with the number of pairs observed, under $\Delta = 0$ and $\Delta = 0.2$.

5.6 Streaming deletion waves

Because deletion requests arrive continually, we test the composition rules of Theorem 6 on histories of 10 waves, and Table 8 bears out both halves of the theorem. When every wave is exactly unlearned, the α-spending per-wave alarm never once fires falsely and the global product alarm raises a false alarm in 2 of 1,000 histories. When a single wave is genuinely bad ($\Delta = 0.25$), its own alarm catches it in 98% of histories and—crucially—the consensus certificate is *never* falsely issued for the history—whereas multiplying certificate e-values falsely certifies that history in *every* run (Table 7), which is the failure Theorem 6(a) predicts. One cost of consensus deserves stating, since it is a headline rule and the number is not flattering: requiring *every* wave to issue means a history of ten *perfectly clean* waves earns a history-wide certificate only 48% of the time at 512 pairs per wave, because each wave independently risks landing in the undetermined band. Consensus buys soundness across waves and pays for it in exactly this way, and a provider auditing a long deletion history should size per-wave cohorts with K in mind rather than sizing one wave and multiplying. The most telling case is the one the product alarm was designed for: 10 waves each leaking only weakly ($\Delta = 0.06$), so weakly that a per-wave test detects them just 14% of the time, are nonetheless caught by the accumulated product alarm in 62% of histories, at a median firing wave of three. Distributed leakage that hides beneath every individual wave's threshold is thus surfaced by composition—a capability no fixed-sample audit possesses.

5.7 End-to-end certification on real models

We now let the abstract null meet a genuine training pipeline. Table 9 reports GPT-2 fine-tuned on a corpus with 640 injected canary pairs and then subjected to each unlearning method. The pattern is what a working certificate should produce: the un-unlearned model is revoked in every seed within about 11 pairs, its canary loss gap averaging 2.2 nats; retraining from a fresh adapter and all three genuine unlearning

Table 7 The negative half of Theorem 6(a), measured: histories of K=10 waves (512 pairs each, 500 histories) containing one genuinely bad wave ($\Delta = 0.25 > \varepsilon = 0.2$), restricted to the histories whose bad wave's *realised* cohort advantage does violate the tolerance. Multiplying certificate e-values—the instinctive move—falsely certifies the whole history every time.

history-wide rule	false certification	95% CI
all-pass consensus (ours)	0.000	[0.000, 0.009]
product of certificate e-values	**1.000**	[0.991, 1.000]

Table 8 Streaming composition over K=10 waves (512 pairs each, 500 histories). The dagger excludes the genuinely bad wave. Bold marks the two payoffs: no false history-wide certificate, and distributed weak leakage caught only by the product alarm.

deletion history	per-wave FWER	global alarm	global cert.
all 10 waves exact	0.000	0.002	0.48
one bad wave ($\Delta = 0.25$)	$0.000^{\dagger}$	0.922	**0.000**
ten weak waves ($\Delta = 0.06$)	0.136	**0.616**	0.000

methods earn certificates in every seed; and the deliberately weakened NPO together with the post-relearning probe sit at the boundary of the undetermined band.

One episode during development illustrates what statistical certification buys beyond a pass/fail label. An early version of our NPO implementation carried a sign error that caused its loss to optimise *toward* memorisation rather than away from it; the revocation arm flagged every such run within roughly 11 pairs, before we had noticed the bug ourselves. A certificate that can catch a defect in the unlearning *code*, not merely in its outcome, is doing more than a descriptive metric can. A CPU-affordable tier built around a small transformer, where retraining is cheap enough to serve as ground truth, reproduces every one of these qualitative outcomes.

5.8 The standard benchmarks: TOFU and MUSE-News

The central empirical test is whether VOUCH behaves correctly on the benchmarks the community actually uses. We inject canaries in each benchmark's native format—question–answer pairs for TOFU, record-style sentences for MUSE-News—into its fine-tuning mixture, and we track held-out utility for every subject. Table 10 and Figures 6 and 7 report verdicts across two benchmarks, three architectures spanning 124M to 1.4B parameters, and 3 seeds each. Table 12 gives every verdict count with a Clopper–Pearson interval, because at $n = 18$ these counts carry wide uncertainty that a bare fraction does not convey.

Detection works, at the stated tolerance. The un-unlearned model is revoked in 16 of 18 benchmark runs under the default sign arm—95% CI $[0.65, 0.99]$—all six Phi-1.5 runs at log-wealth between 149 and 186. The two remaining runs are the two weakest-leaking GPT-2 cells, and here we must report something the earlier version

Table 9 GPT-2 end-to-end over 2 seeds (640 pairs, two-sided $\varepsilon = 0.2$, cohort target): per-seed verdicts (**I** issued, **R** revoked, **U** undetermined), the mean confidence-sequence upper bound on Δ, and the mean canary score gap $\bar{D}$ with its across-seed standard deviation. Seeds 1 and 2 are shown; the seed-0 run of this tier predates per-pair logging and cannot be replayed under the cohort target, so it is omitted rather than mixed in from the earlier analysis.

subject	verdicts	mean U_t on Δ	mean $\bar{D}$	sd $\bar{D}$
no unlearning	**R**/**R**	0.90	+2.23	0.12
retrain (fresh adapter)	**I**/**I**	0.14	-0.06	0.04
GA	**I**/**I**	0.21	+0.07	0.04
GradDiff	**I**/**I**	0.14	-0.19	0.10
NPO	**I**/**I**	0.15	+0.03	0.04
NPO (25% budget)	**I**/**I**	0.22	+0.29	0.13
NPO + P1 relearn	**I**/**I**	0.22	+0.31	0.02
NPO + P2 4-bit	**I**/**I**	0.14	-0.00	0.04
NPO + P3 jailbreak	**I**/**I**	0.16	+0.06	0.03

did not surface: under the corrected, more powerful cohort target those two runs are not merely undetermined, they are *certified*. Their realised advantages are 0.062 and 0.078, genuinely below the $\varepsilon = 0.2$ tolerance, so a certificate is the correct verdict for the claim being made. This is the clearest possible illustration of why the tolerance is not a free parameter: at $\varepsilon = 0.2$ a model that was never unlearned at all can pass, because $\varepsilon = 0.2$ permits an adversary in the declared class to win 60% of the time. We return to how ε should be chosen at the end of this subsection.

Exact unlearning certifies. Retraining from a fresh adapter is issued on all 18 runs at $\varepsilon = 0.2$ and on all 18 at $\varepsilon = 0.1$; at $\varepsilon = 0.05$ it is issued on 12 of 18 (Table 11). This is the end-to-end result at a tight tolerance that was missing from the previous version, and it comes with a caveat that must travel with it: it is a *cohort* certificate. On the super-population target, no run certifies at $\varepsilon = 0.05$ at these cohort sizes, and Theorem 4 says none should—$\varepsilon = 0.05$ needs roughly 2,400 pairs and we plant at most 640. Both columns are in Table 11 and we recommend reading them together.

The certificate/utility pairing exposes destructive unlearning. Gradient ascent earns forgetting certificates while its held-out loss balloons to between 58 and 286 nats per seed—the cell means in Table 10 run from 60 to 179—against retraining's 1.5 to 3.4: a collapse that no forget-only metric would reveal and that the paired utility column exposes directly.

A reviewer observed, correctly, that held-out NLL identifies severe collapse but does not establish general utility, so we added the benchmark's own forget and retain readings and a general-knowledge probe (Table 13). They sharpen rather than soften the picture. Gradient ascent reaches a forget NLL of 94.3 and a retain NLL of 93.7 with ROUGE-L recall of exactly 0.00 on both splits: it has not forgotten the forget set, it has stopped producing language. GradDiff is subtler and more interesting—its retain NLL is 6.8 but its capability NLL is 19.4, so the damage is nearly 3 times larger off the retain distribution than on it, which a retain-only utility reading would have missed entirely. SimNPO [4], added at revision, achieves the widest forget/retain separation of any subject that remains fluent (6.20 against 2.62, versus NPO's 5.50 against 2.60) at a capability NLL of 4.17 against retraining's 3.23, and is the best-behaved of the approximate methods on this axis.

One cell of that table is deliberately blank. For TOFU the P1 relearn corpus *is* the capability probe set (`world_facts`), so a capability reading after P1 measures training on the probe rather than retained ability; it reads 1.46, better than the retrained reference, which is an artefact and not a finding. We report it as unavailable rather than quietly printing a flattering number, and note that disentangling the two would require a probe set disjoint from the relearn corpus.

Two-sided certification matters, though not on these runs. On several TOFU seeds gradient ascent, GradDiff and NPO push the canaries *below* chance, with mean score gaps as negative as −0.54 (GradDiff on TOFU/Pythia) and −0.51 (NPO on TOFU/Phi-1.5). We should be exact about what follows from that, because an earlier draft was not: at $\varepsilon = 0.2$ these gaps are still inside the tolerance, so the two-sided arm issues here too and nothing is withheld on any TOFU cell. What the two-sided arm buys is that the *same* runs would be certified by a one-sided test at *any* tolerance, however tight, because a one-sided test cannot see below chance at all; the two-sided arm prices them correctly, and at $\varepsilon = 0.05$ 5 of the 6 retrain runs that fail to issue are held back by the *lower* arm, not the upper one—that is, by over-forgetting rather than by residual memorisation. The behavioural signature is the one Yuan et al. [7] and Li et al. [9] describe from the optimisation side, showing up in an audit.

Recoverability is measured rather than argued. At $\varepsilon = 0.2$ on the cohort target the relearn probe leaves NPO's certificate intact on all three TOFU architectures and on MUSE/Pythia; it *revokes* on one MUSE/GPT-2 seed and on one MUSE/Phi-1.5 seed. So on this benchmark tier the probe changes the verdict in exactly 2 of 18 runs, both on MUSE-News—the more open-domain corpus, and the one where the public relearn set sits closest in distribution to the forget set. Recoverability is benchmark- and model-dependent, exactly as Hu et al. [8] report, and VOUCH quantifies the dependence rather than assuming it away. But we should be plain that at $\varepsilon = 0.2$ the probe is close to inert on our runs: 16 of 18 certificates survive it, and the two that do not are revocations rather than downgrades. A relearn probe becomes informative as the tolerance tightens, which is another reason not to read $\varepsilon = 0.2$ as a recommendation.

Underlying all of this is a clean dose–response signal, measured on the *un-unlearned* models, where there is memorisation to dose. Mean score gaps rise monotonically with training-time repetition—+0.46, +0.78, +1.54 and +2.35 nats at repetition factors 1, 2, 4 and 8, averaged over the six benchmark/model cells—and so do the realised advantages, +0.22, +0.35, +0.59 and +0.73. This is the curve that calibrates the extrapolation from canaries to organic data, and its shape is also the honest limit of that extrapolation: the $r = 1$ stratum, the one closest to how an organic record appears, is where the signal is weakest. Figure 5 makes the whole mechanism concrete on a single real canary pair.

A certificate and its confidence sequence can disagree, and often do. A reader comparing Tables 10 and 17 will notice issued certificates sitting beside confidence-sequence upper bounds that exceed the very tolerance they certify: across the benchmark tier this happens in 46 of 107 issued runs, with U_t reaching 0.232 against

Table 10 Benchmark verdicts per seed (sign-based arm, cohort target, $\varepsilon = 0.2$) and mean held-out utility (NLL), across three architectures. TOFU on GPT-2 and Pythia uses 384 pairs; TOFU on Phi-1.5 and all MUSE-News runs use 512. 3 seeds per cell.

	TOFU/GPT-2		TOFU/Pythia		TOFU/Phi-1.5		MUSE/GPT-2		MUSE/Pythia		MUSE/Phi-1.5	
subject	verdicts	util.	verdicts	util.	verdicts	util.	verdicts	util.	verdicts	util.	verdicts	util.
no unlearning	R/R/I	2.2	R/R/R	1.9	R/R/R	1.5	R/R/I	3.3	R/R/R	3.4	R/R/R	3.3
retrain (fresh adapter)	I/I/I	2.0	I/I/I	1.8	I/I/I	1.5	I/I/I	3.2	I/I/I	3.4	I/I/I	3.4
GA	I/I/I	87.2	I/I/I	170.4	I/I/I	100.7	I/I/I	59.8	I/I/I	179.1	I/I/I	90.1
GradDiff	I/I/I	6.0	I/I/I	10.5	I/I/I	7.8	I/I/I	4.2	I/I/I	4.4	I/R/I	3.9
NPO	I/I/I	2.6	I/I/I	2.2	I/I/I	1.6	I/I/I	3.3	I/I/I	3.6	I/I/I	3.4
NPO + P1 relearn	I/I/I	2.8	I/I/I	2.6	I/I/I	2.2	I/I/R	3.2	I/I/I	3.4	I/R/I	3.3
NPO + P3 jailbreak	I/I/I	2.6	I/I/I	2.2	I/I/I	1.6	I/I/I	3.3	I/I/I	3.6	I/I/I	3.4

$\varepsilon = 0.2$. This is not an inconsistency in the sense of one object being wrong, and it is worth explaining because it looks like one. The two arms are different procedures with different budgets: the certificate is a sequential equivalence test that rejects at $1/\alpha$, while the confidence sequence intersects two-sided wealth processes at $2/\alpha$ per grid point, and the latter is a strictly more demanding object—it must cover the truth at every time and for every candidate value simultaneously. Each is separately valid, and a procedure that rejects the null $|\Delta| \geq \varepsilon$ need not also produce an interval contained in $(-\varepsilon, \varepsilon)$. A verifier who wants the stronger, self-evidently coherent statement can define issuance as "$U_t < \varepsilon$" instead; that is a valid rule, it implies the certificate, and it costs roughly a factor of two in pairs. We report the certificate as the headline verdict because it is the object the theory is stated for, and we print U_t alongside because a bounded quantity is more useful to an auditor than a bare verdict—but the two should be read as two answers to two questions, not as one answer checked twice.

Ties. Exact ties $D = 0$ occur in 0 of 124,416 scored pairs for s_{loss} across every architecture. The one exception is s_{mink} on Gemma-4-E2B, where 2.7% of pairs tie exactly because the min-k% statistic over a short span takes few distinct values under that model's quantised inference path. The committed tie-break coin is therefore not a formality in that cell, and it is what keeps Theorem 1 exact there.

Choosing ε. The tolerance has no universal value, and we would rather give a defensible procedure than pretend otherwise. ε is the residual advantage a declared adversary is permitted, so $\varepsilon = 0.2$ licenses a 60% distinguishing rate and $\varepsilon = 0.05$ a 52.5% one. Three considerations set it in practice. The regulatory one: if the audit must support a claim that erasure is demonstrable under an accountability duty, the tolerance should be small enough that the permitted advantage is not itself reportable as a privacy loss, which argues for $\varepsilon \leq 0.05$. The statistical one: Theorem 4 fixes the cohort at roughly $2\log(1/\alpha)/\varepsilon^2$ pairs, so ε is bought with canaries—600 at 0.1, 2,400 at 0.05, 15,000 at 0.02. The operational one: the cohort must stay a negligible share of the corpus, which for a provider with millions of records makes even $\varepsilon = 0.02$ affordable and for a benchmark-scale corpus does not. Our own headline $\varepsilon = 0.2$ is a demonstration tolerance chosen so that 384–640 pairs suffice on academic-scale corpora; it is not a recommendation, and we say so wherever the number appears.

A worked canary pair (TOFU/Phi-1.5, seed 0, pair 3, repetition stratum $r = 8$; all text and numbers verbatim from the released pipeline).

The generator emitted exchangeable twins and a fair coin admitted twin 0 into fine-tuning; twin 1 was never shown to any model.

in-twin	`Question: What is the confidential badge identifier of Alaric Nightingale? Answer: SUUHWXGI1F`
ghost twin	`Question: What is the confidential badge identifier of Alaric Ivarsson? Answer: 8247TLTKPP`

Asking the *in-twin* question and decoding greedily:

model	*answer*	*secret NLL (in / ghost)*
fine-tuned	`7E5SU30039A0L8WL`	0.99 / 9.03; $D = +8.0$
NPO-unlearned	`LLLLLLLLLLLLLL`	77.2 / 81.7; $D = +4.5$
retrained	"Alaric Nightingale is known for his unique identification badge, . . . "	8.26 / 9.86; $D = +1.6$

Recall the sign convention of Section 3.4: the score is the token-normalised *log-likelihood*, so D equals the ghost's secret NLL minus the in-twin's and a *positive* D means the trained twin is better remembered. The fine-tuned model does not reproduce the ten-character secret verbatim under greedy decoding, yet it assigns it 0.99 nats per token against 9.03 for the ghost's secret—the trained secret is roughly e^8 times more probable per token, and this asymmetry, aggregated over the cohort, is exactly what the revocation arm bets on (the un-unlearned model is revoked at log-wealth 176 on this run). After NPO unlearning the model produces degenerate text *on the canary prompts specifically*—held-out utility is untouched (Table 10)—and its secret NLLs are large on *both* twins; the residual gap of +4.5 flips sign on the very next pair, single-pair noise once memorisation is gone, which is why the verifier bets over hundreds of coin-randomised pairs rather than trusting any one of them. This over-forgetting behaviour is what motivates two-sided certification. The retrained model answers fluently, knows nothing, and treats the twins nearly symmetrically; its certificate is issued at pair 220 on the cohort target and 258 on the super-population one.

Fig. 5 One canary pair, end to end: what each model answers and the score gap D that the betting processes consume. With scores oriented so that higher means more memorised (Section 3.4), D is the ghost-secret NLL minus the in-secret NLL, and $D > 0$ is the direction of leakage.

5.9 A subject that never touches the token loss: RMU

Every subject so far optimises a token-level objective, and a reviewer reasonably asked whether the framework's conclusions survive a method of a different kind. We therefore add *representation misdirection* [5], the method introduced with WMDP, which acts in representation space rather than on the loss: at one hidden layer it pushes the forget set's activations toward a fixed random direction while holding the retain set's activations near the frozen reference model's. Nothing in VOUCH changes to accommodate it—the audit queries M_u as a black box—which is the point of testing it.

Table 14 reports RMU beside the two other utility-preserving subjects and the two controls, all co-trained in a single run on TOFU/GPT-2 over the same three seeds as the main benchmark tier, so that every row shares one fine-tuned adapter, one canary cohort and one machine per seed. RMU behaves as a genuine unlearning method. It

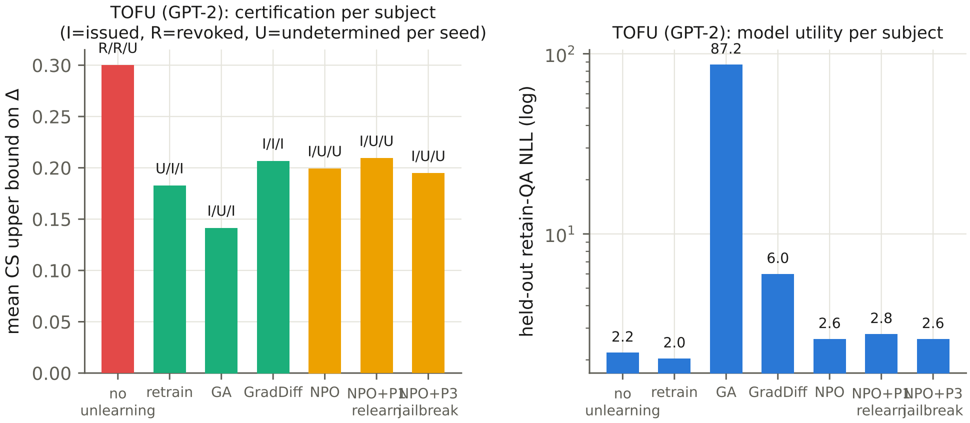

Fig. 6 TOFU (GPT-2). Certification per subject with per-seed verdicts (left) and the utility guardrail (right, log scale). Gradient ascent earns its certificate while destroying held-out performance.

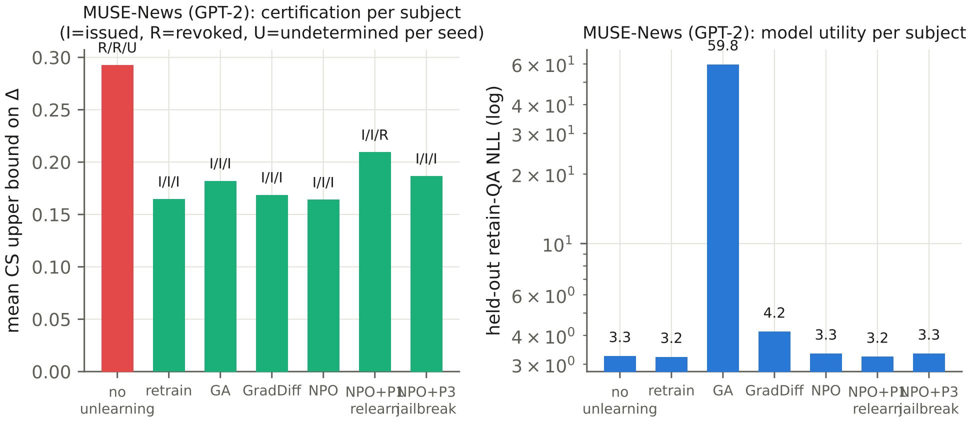

Fig. 7 MUSE-News (GPT-2, 512 pairs). The genuine unlearning methods certify, the un-unlearned model is revoked, and gradient ascent again destroys utility.

certifies at $\varepsilon = 0.2$ on all three seeds, and its realised cohort advantage of $+0.050$ is close to what retraining a fresh adapter achieves on the same cohorts ($+0.026$), against the un-unlearned model's $+0.097$. It does that at a retain NLL of 2.55 against retraining's 1.99, so it is a genuine member of the utility-preserving family rather than destructive unlearning in disguise.

The interesting comparison is with the other two, and it is a distinction the certificate is built to draw. NPO and SimNPO buy their forgetting by raising the forget split's likelihood a long way—to 6.34 and 5.79 against the un-unlearned model's 2.19—while RMU raises it only to 2.73, barely moving it. Yet all three end up at a comparable

Table 11 Tolerance sweep for exact unlearning (retraining from a fresh adapter), 3 seeds per cell. At $\varepsilon = 0.05$ the cohort target issues on 12 of 18 runs and on all three seeds in two of the six benchmark/model cells; the super-population target issues on none at $\varepsilon \leq 0.1$, as Theorem 4 predicts at these cohort sizes.

		cohort target			super-population target		
benchmark / model	m	ε=0.2	0.1	0.05	0.2	0.1	0.05
TOFU/GPT-2	384	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**I**/**I**	**U**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**
TOFU/Pythia	384	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**
TOFU/Phi-1.5	512	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**
MUSE/GPT-2	512	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**U**/**I**	**I**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**
MUSE/Pythia	512	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**U**/**I**	**I**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**
MUSE/Phi-1.5	512	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**U**/**U**	**I**/**I**/**I**	**U**/**U**/**U**	**U**/**U**/**U**

Table 12 Verdict counts pooled over the six benchmark/model cells with Clopper–Pearson 95% intervals ($n = 18$).

subject	target verdict	count	95% CI
no unlearning	revoked	16/18	[0.65, 0.99]
retrain (fresh adapter)	issued	18/18	[0.81, 1.00]
GA	issued	18/18	[0.81, 1.00]
GradDiff	issued	17/18	[0.73, 1.00]
NPO	issued	18/18	[0.81, 1.00]
NPO + P1 relearn	issued	16/18	[0.65, 0.99]
NPO + P3 jailbreak	issued	18/18	[0.81, 1.00]

place on the quantity the certificate measures (−0.014, −0.035 and +0.050). A forget-quality metric reading likelihoods would therefore rank RMU far below the other two; the certificate treats them as comparable, because what it measures is whether the deployed model still lets an attacker *distinguish* a trained twin from its ghost. Which ranking is right depends on what the audit is for, and this is the clearest instance in the study of the two readings coming apart on real models.

The un-unlearned model is the one row where the three seeds disagree—revoked on two, issued on the third with a realised advantage of +0.042, comfortably below $\varepsilon = 0.2$—and we read that as informative rather than as noise to average away. It is the same phenomenon Table 10 already documents for two cells of the main benchmark tier: $\varepsilon = 0.2$ permits enough residual advantage that an un-unlearned model can certify on a training run where the canaries happened to be memorised weakly. A positive control that fails once in three seeds is not a control that has stopped working; it is a measurement of how loose the demonstration tolerance is, on exactly the axis Section 5.8 already reports it.

Table 13 Benchmark forget/retain readings and a general-knowledge probe, TOFU/GPT-2, 3 seeds, $\varepsilon = 0.2$ on the cohort target. NLL is the mean over the benchmark's own splits; ROUGE-L recall is greedy-decode against the reference continuation, the TOFU/MUSE reporting convention. ‡For TOFU the P1 relearn corpus is the capability probe set, so that cell is contaminated by construction and is not reported.

		mean NLL			ROUGE-L recall	
subject	verdicts	forget	retain	capability	forget	retain
no unlearning	**R**/**R**/**I**	2.19	2.14	3.22	0.32	0.33
retrain (fresh adapter)	**I**/**I**/**I**	2.53	1.98	3.23	0.31	0.34
GA	**I**/**I**/**I**	94.31	93.71	86.65	0.00	0.00
GradDiff	**I**/**I**/**I**	7.98	6.79	19.44	0.24	0.29
NPO	**I**/**I**/**I**	5.50	2.60	3.84	0.20	0.26
SimNPO	**I**/**I**/**I**	6.20	2.62	4.17	0.17	0.28
NPO + P1 relearn	**I**/**I**/**I**	3.72	2.70	n/a‡	0.11	0.14
NPO + P3 jailbreak	**I**/**I**/**I**	5.50	2.60	3.84	0.20	0.26

Table 14 Representation misdirection (RMU) beside the other utility-preserving subjects, co-trained on TOFU/GPT-2 with a shared fine-tuned adapter, 384 pairs, cohort target. Verdicts are per seed (**I** issued, **R** revoked, **U** undetermined); $\hat{\Delta}$ is the realised cohort advantage of s_{loss}; forget and retain NLL are on the benchmark's own splits. It uses the same protocol, cohort size, tolerances and seed count as the benchmark tier of Section 5.8, and is reported as its own tier because its subjects were co-trained in a single run.

	verdict				mean NLL	
subject	ε=0.2	0.1	0.05	$\hat{\Delta}$	forget	retain
no unlearning	**R**/**R**/**I**	**R**/**R**/**U**	**R**/**R**/**U**	+0.097	2.19	2.14
retrain (fresh adapter)	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**I**/**I**	+0.026	2.53	1.99
NPO	**I**/**I**/**I**	**I**/**I**/**I**	**I**/**U**/**U**	-0.014	6.34	2.83
SimNPO	**I**/**I**/**I**	**I**/**I**/**I**	**U**/**U**/**U**	-0.035	5.79	2.43
RMU	**I**/**I**/**I**	**I**/**U**/**I**	**U**/**U**/**U**	+0.050	2.73	2.55

5.10 RMU on a second architecture

Everything in Section 5.9 is on one architecture, so a natural question is whether the RMU tier's pattern is a property of GPT-2 or of the method. Table 15 repeats the identical protocol—the same three seeds, cohort size, tolerances and co-training design—on TOFU/Pythia-160M. The pattern transfers. RMU again certifies at $\varepsilon = 0.2$ and $\varepsilon = 0.1$ on all three seeds and moves to undetermined at $\varepsilon = 0.05$ on all three, the same tolerance at which it stops certifying on GPT-2; its realised advantage of +0.026 again sits close to retraining's +0.014, both well below the un-unlearned model's +0.535. It again does this while barely moving the forget-split likelihood—to 9.95 nats against the un-unlearned model's 1.78, a far smaller shift than NPO's 76.73 or SimNPO's 65.65 on the same seeds—so the loss/detectability divergence Section 5.9 reports is not an artifact of one model family.

Table 15 The RMU tier of Table 14, repeated unchanged on TOFU/Pythia-160M (three seeds, 384 pairs, cohort target). Same columns as Table 14.

	verdict				mean NLL	
subject	ε=0.2	0.1	0.05	$\hat{\Delta}$	forget	retain
no unlearning	**R/R/R**	**R/R/R**	**R/R/R**	+0.535	1.78	1.89
retrain (fresh adapter)	**I/I/I**	**I/I/I**	**I/I/U**	+0.014	2.54	1.80
NPO	**I/I/I**	**I/U/I**	**U/U/I**	+0.061	76.73	2.05
SimNPO	**I/I/I**	**I/I/I**	**U/U/I**	+0.005	65.65	3.01
RMU	**I/I/I**	**I/I/I**	**U/U/U**	+0.026	9.95	3.84

The positive control is if anything sharper here: the un-unlearned model is revoked on all three Pythia seeds at $\varepsilon = 0.2$, rather than on two of three as on GPT-2, so the one seed where the GPT-2 control slipped through is a property of that particular training run rather than of the protocol.

5.11 A tight-tolerance cohort, and the limit it exposes

The tolerance sweep of Table 11 reaches $\varepsilon = 0.05$ on the cohort target but nothing below $\varepsilon = 0.2$ on the super-population one, and Theorem 4 says why: $\varepsilon = 0.1$ needs about 600 pairs and $\varepsilon = 0.05$ about 2,400, against the 384–640 we plant. So we planted a large cohort and asked what happens. Table 16 reports TOFU/GPT-2 with 3,072 pairs, 2 seeds, and—this matters for the reading—*every canary inserted exactly once*, $r = 1$, which keeps the corpus share at 22.5% and puts the audit in the most organic-like stratum.

Two results and one caveat. First, the super-population target now certifies at $\varepsilon = 0.1$: all four subjects, both seeds. That is the reviewers' request met on the target they asked about, and it lands exactly where Theorem 4 predicts it should once the cohort clears 600 pairs. Second, at $\varepsilon = 0.05$ the super-population arm is still undetermined at 3,072 pairs. The asymptotic figure of 2,400 is for a one-sided crossing under exact unlearning; two-sided certification costs about a factor of 1.7 (Section 5.5), so $\varepsilon = 0.05$ wants something closer to 4,000 pairs, and we do not have it. The cohort target certifies at $\varepsilon = 0.05$ on all four subjects.

The caveat is the more instructive finding, and it cuts against the instrument rather than the inference. At $r = 1$ this model memorises essentially nothing: the realised advantages in Table 16 span -0.040 to $+0.025$, and the mean score gap for the *un-unlearned* model is $+0.005$ nats. So "no unlearning" certifies at $\varepsilon = 0.05$ alongside retraining—correctly, because on this cohort there is no residual advantage to find. A single exposure of a single canary in a 6,872-document corpus leaves no measurable trace in a 124M-parameter model, which is a statement about GPT-2 and about single-exposure memorisation, not about VOUCH. Compare the same stratum in the repetition-stratified runs, where $r = 1$ carries $+0.122$ nats on TOFU/GPT-2 and $+1.244$ on TOFU/Phi-1.5: the signal at $r = 1$ grows with model capacity, and it grows when the corpus also contains repeated copies of *other* canaries from the same template.

Table 16 Tight-tolerance cohort: TOFU/GPT-2, 3,072 canary pairs, every canary inserted once ($r = 1$), 2 seeds, 22.5% canary character share. The super-population target reaches $\varepsilon = 0.1$ here, which it does not at the 384–640-pair cohorts of Table 11. At $r = 1$ this model memorises so little that the un-unlearned model also certifies: $\hat{\Delta}$ is the realised cohort advantage per seed.

	cohort target			super-population target			
subject	ε=0.2	0.1	0.05	0.2	0.1	0.05	$\hat{\Delta}$
no unlearning	**I/I**	**I/I**	**I/I**	**I/I**	**I/I**	**U/U**	+0.025, -0.005
retrain (fresh adapter)	**I/I**	**I/I**	**I/I**	**I/I**	**I/I**	**U/U**	-0.005, -0.012
NPO	**I/I**	**I/I**	**I/I**	**I/I**	**I/I**	**U/U**	-0.010, +0.008
SimNPO	**I/I**	**I/I**	**I/I**	**I/I**	**I/I**	**U/U**	-0.017, -0.040

This is the sharpest form of the canary-to-organic caveat we can state, and it is sharper than the dose–response curve alone conveys. The audit's power at $r = 1$ is not merely lowest; on GPT-2 it is nil. An organic forget record seen once by a small model may simply not be memorised, in which case there is nothing for any behavioural audit to certify the removal of—and a certificate issued in that regime is uninformative rather than wrong. Auditing at a tolerance tight enough to be operationally meaningful therefore requires either a model large enough to memorise single exposures, or canaries dosed above the organic rate, and the second is the choice we made for the rest of this study. We regard closing that gap—power at $r = 1$ on models large enough for it to matter—as the most important open problem this framework faces.

5.12 Head to head against the descriptive metrics

The paper's motivating claim is that descriptive metrics are inadequate, and a reviewer rightly observed that we had asserted rather than shown it. We therefore run, on the same models and the same seeds, two metrics standing in for the field's conventions: a TOFU-style forget-quality analogue, a two-sample Kolmogorov–Smirnov test between the unlearned and the retrained model's canary-score distributions, whose convention is that a *large* p-value means "indistinguishable from retrained, therefore forgotten"; and a MUSE-style min-k% leakage analogue, the paired AUC expressed relative to the retrained reference as PrivLeak does. Table 17 reports them beside VOUCH's verdicts.

Three disagreements are worth naming. First, the forget-quality analogue is *systematically* pessimistic on this data: it fails almost every genuine unlearning method on Pythia and Phi-1.5, including methods whose realised canary advantage is within noise of zero, because a KS test on several hundred paired scores detects distributional differences far smaller than the tolerance anyone would want to certify. It answers "are these distributions identical" when the operative question is "is the residual advantage below ε," and those are different questions. Second, PrivLeak reads -2.4% on the single MUSE/Phi-1.5 GradDiff seed that VOUCH *revokes*—a *negative* leakage figure on a model carrying residual memorisation strong enough to trip an anytime alarm, and -1.2% averaged over that cell's 3 seeds, which is the number the table shows. This is the concrete instance of the claim, and we flag that it rests on one seed. Third, and

Table 17 VOUCH verdicts beside descriptive metrics on the same runs and seeds. "forget-KS" is the TOFU-style forget-quality analogue (P = pass, F = fail, per seed); "PrivLeak" the MUSE-style min-k% leakage relative to the retrained reference; "min-k peek" whether a min-k% test inspected after every pair ever flags leakage (L) or not (.).

		VOUCH		descriptive metrics		
benchmark / model	subject	verdicts	$U_t(\Delta)$	forget-KS	PrivLeak	min-k peek
TOFU/GPT-2	no unlearning	**R/R/I**	0.17	F/F/F	+13.8%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.13	P/P/P	+0.0%	L/./L
	GradDiff	**I/I/I**	0.19	P/P/P	-2.2%	././.
	NPO	**I/I/I**	0.16	P/P/F	+2.0%	./L/L
TOFU/Pythia	no unlearning	**R/R/R**	0.53	F/F/F	+52.5%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.12	P/P/P	+0.0%	./L/.
	GradDiff	**I/I/I**	0.16	F/F/F	-2.7%	././.
	NPO	**I/I/I**	0.17	F/F/F	-1.3%	./L/.
TOFU/Phi-1.5	no unlearning	**R/R/R**	0.77	F/F/F	+71.8%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.18	P/P/P	+0.0%	././.
	GradDiff	**I/I/I**	0.20	F/F/F	+0.5%	././.
	NPO	**I/I/I**	0.20	F/F/F	+1.8%	L/./L
MUSE/GPT-2	no unlearning	**R/R/I**	0.19	F/F/F	+17.0%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.17	P/P/P	+0.0%	././.
	GradDiff	**I/I/I**	0.17	F/F/F	+1.6%	././.
	NPO	**I/I/I**	0.14	F/F/F	+3.1%	././L
MUSE/Pythia	no unlearning	**R/R/R**	0.55	F/F/F	+50.4%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.13	P/P/P	+0.0%	././.
	GradDiff	**I/I/I**	0.21	F/F/F	-0.8%	././L
	NPO	**I/I/I**	0.20	F/F/F	+1.3%	./L/.
MUSE/Phi-1.5	no unlearning	**R/R/R**	0.78	F/F/F	+79.1%	L/L/L
	retrain (fresh adapter)	**I/I/I**	0.17	P/P/P	+0.0%	L/./.
	GradDiff	**I/R/I**	0.16	F/F/F	-1.2%	././.
	NPO	**I/I/I**	0.21	F/F/F	+3.0%	./L/.

most directly on the paper's thesis, the min-k% statistic inspected after every pair—the way an auditor with a stream of deletions would naturally use it—flags leakage on genuinely clean retrained models in several cells, which is optional stopping doing exactly what Section 5.2 predicts.

We should be even-handed about what this does not show. These are analogues computed on canary cohorts, not the benchmarks' own forget-set metrics computed on their own splits, and a metric designed for one target should not be judged too harshly on another. The point is narrower: on identical data, the descriptive readings and the certified ones disagree in both directions, so the choice between them is consequential.

5.13 Attacks from outside the declared class

The certificate is a supremum over $\mathcal{F}$, so a stronger attack outside $\mathcal{F}$ could exceed ε on a certified model without contradicting anything we prove. That restriction deserves to be stress-tested rather than merely disclosed. We run two attacks that no member of $\mathcal{F}$ implements, against every model VOUCH certified at $\varepsilon = 0.2$ for which the attack is defined (178 model/subject/seed combinations). The first is *stratum-restricted*: the repetition label r_i is part of the committed manifest, so a verifier may condition on it, and restricting the audit to the $r = 8$ stratum concentrates it on the most-memorised pairs. The second is a *cross-fitted learned combination*: a Fisher direction over the declared scores estimated on the first half of the reveal order and applied to the second. Both keep Theorem 1 intact—the first conditions only on committed per-pair data, the second uses a predictable weight vector—so both come with anytime-valid bounds rather than point estimates.

Table 18 reports the outcome, and it is a *null* result. The stratum-restricted attack exceeds $\varepsilon = 0.2$ on 2.8% of certified runs and the learned combination on 0.5%. Those look like breaches until one asks what a *clean* subsample of the same size would do: the attacks are evaluated on 96–159 and 192–1,536 pairs respectively, and an exact binomial calculation puts the chance exceedance rate at 3.5% and 0.3%. The stratum-restricted attack therefore exceeds the tolerance *less* often than noise predicts, and the learned combination's single exceedance in 184 runs is within a coin-flip of its own null. The anytime confidence bounds tell the same story: their means, 0.078 and 0.063, sit far below ε.

So this experiment does not show what we had hoped it would show in either direction. It does not demonstrate that $\mathcal{F}$ is too narrow, because the two attacks we could build produce no signal beyond subsampling noise. And it does not license the converse—that $\mathcal{F}$ is wide enough—because both attacks are weak: one is a subpopulation restriction that pays for its focus in sample size, the other a linear recombination of scores already in $\mathcal{F}$. The honest position is that the certificate's dependence on $\mathcal{F}$ is a *definitional* limit, visible in Theorem 2's statement rather than in these two measurements. Putting a number on it needs an attack that demonstrably works where leakage is present, which is what Section 5.14 supplies.

We therefore state the guarantee as what it is—a statement about $\mathcal{F}$—rather than as a bound on the largest residual advantage any score whatever could extract. What the design does offer is that enlarging $\mathcal{F}$ costs power and no validity—any score a sceptic names can be added, and Theorem 2 needs no multiplicity correction—so the framework absorbs a stronger attack without any change to the theory. Two observations on where such an attack would come from. The subpopulation reading is worth keeping in mind independently of this table: the certificate bounds a cohort *average*, and our own dose–response curve reports realised advantages of $+0.73$ at $r = 8$ against $+0.22$ at $r = 1$, so an average below the tolerance is compatible with a stratum above it. Certifying per stratum removes that gap at the cost of multiplying the cohort requirement by the number of strata, and we report the trade-off rather than silently adopting either default. And the most promising direction is a paraphrase-aware score, since Li et al. [9] show that gradient-ascent objectives move probability mass onto

Table 18 Attacks from outside the declared class $\mathcal{F}$, run against every model/subject/seed combination certified at $\varepsilon = 0.2$ for which the attack is defined. "observed" is the fraction of runs on which the attack's realised advantage exceeds ε; "null" is what an exact binomial calculation predicts for a *clean* subsample of the same size, averaged over the run-specific sizes. Neither attack beats its own null, so the table is a null result: it bounds neither how narrow $\mathcal{F}$ is nor how wide. Bold marks an observed rate above the null.

			$\lvert\hat{\Delta}\rvert > \varepsilon$		anytime CS	
attack	runs	pairs	observed	null	mean U_t	exceed
stratum-restricted ($r = 8$ only)	178	$96 - -159$	0.028	0.035	0.078	0.028
cross-fitted learned combination	184	$192 - -1536$	**0.005**	0.003	0.063	0.005

semantically equivalent rephrasings—mass that our literal-span scores cannot see by construction.

The strongest comparison a sceptic would want is a full LiRA-style attack with shadow models. It needs the whole pipeline repeated for every shadow, which the tiers in this section cannot afford; Section 5.14 runs it on the tier that can, and reports both what it finds and how far its scope reaches.

5.14 A shadow-model attack from outside $\mathcal{F}$

The two attacks of Section 5.13 were weak by construction, and their null result was correspondingly uninformative: neither beat the noise floor of subsampling, so neither bounded the gap between $\sup_{s \in \mathcal{F}} \lvert\Delta^{(s)}\rvert$ and what a determined attacker could extract. The attack a sceptic actually wants is LiRA [19], which calibrates the target model's confidence on each example against the distribution of confidences that *shadow* models trained with and without that example produce. LiRA is outside $\mathcal{F}$ in a strong sense: every score in $\mathcal{F}$ is a function of the deployed model's outputs alone, whereas LiRA consumes N additional trained models, so no choice of Q and no reweighting of members of $\mathcal{F}$ reproduces it. It is also not a candidate for inclusion in $\mathcal{F}$, because a verifier operating under VOUCH's cost model—$2Qm$ forward passes and no training—cannot run it.

Running it requires repeating the entire pipeline $N + 1$ times, which the 124M–5.1B tiers cannot afford on the compute available here. We therefore run it on the TinyGPT tier, the one this study already uses where retraining is cheap enough to serve as ground truth (Section 5.7), with 16 shadows and a cohort of 256 pairs. Each shadow repeats the whole pipeline on the same twin pool with an independent redraw of the inclusion coins, so every twin is trained on in about half the shadows and withheld in the rest, and the unlearned subject is calibrated against shadows that were themselves unlearned rather than against a pre-unlearning proxy. For twin x we take $\phi(M, x)$ to be the token-normalised log-likelihood of its secret, fit $\mathcal{N}(\mu_{\text{in}}(x), \sigma^2)$ and $\mathcal{N}(\mu_{\text{out}}(x), \sigma^2)$ with a variance pooled across twins—the global-variance variant, which is what a modest shadow count supports—and score $\Lambda(x) = \log \mathcal{N}(\phi; \mu_{\text{in}}, \sigma^2) - \log \mathcal{N}(\phi; \mu_{\text{out}}, \sigma^2)$. The attack guesses the twin with the larger Λ, and its realised cohort advantage is directly comparable to the tolerance.

Table 19 A LiRA-style shadow-model attack from outside $\mathcal{F}$ (TinyGPT tier, 16 matched shadows, cohort $m = 256$, one target seed). "declared $\mathcal{F}$" is the advantage the audit's own score class achieves on the same cohort; "agreement" is the fraction of pairs on which LiRA and the declared class make the same guess. Intervals are Clopper–Pearson on the sign rate: the betting confidence sequence collapses onto the realised value once the cohort is exhausted (Theorem 2), so it carries no width here. The first row is the positive control that makes the second interpretable.

	declared $\mathcal{F}$	LiRA (outside $\mathcal{F}$)			
target model	$\hat{\Delta}$	$\hat{\Delta}$	95% CI	agreement	$\lvert\hat{\Delta}\rvert > \varepsilon$?
no unlearning (positive control)	+0.867	+0.820	[+0.736, +0.884]	0.92	**yes**
NPO, certified at $\varepsilon = 0.2$	+0.047	-0.016	[−0.141, +0.110]	0.47	no

Table 19 reports the outcome, and the positive control is what makes it readable. On the un-unlearned model LiRA achieves $\hat{\Delta} = +0.820$ (95% CI $[+0.736, +0.884]$) and agrees with the declared class's own verdict on 92% of pairs: the attack is potent, and it is seeing the same memorisation the audit sees. On the NPO model that VOUCH certifies at $\varepsilon = 0.2$, the same attack yields $\hat{\Delta} = -0.016$ with 95% CI $[-0.141, +0.110]$, and its agreement with the in-class score falls to 0.47—chance, which is what two statistics guessing independently look like. The interval excludes every tolerance we certify at, so on this model the strongest attack we could mount does not merely fail to beat a noise floor: it is bounded away from ε.

We are careful about what that licenses. This is one target model on one tier, with 16 shadows against the hundreds a canonical LiRA would use, and the certified subject is a single unlearning method. It does not show that $\mathcal{F}$ is wide enough in general, and a larger model, a stronger attack, or a different subject could still exceed the tolerance. What it does do is change the character of the evidence. The attacks in Section 5.13 could not distinguish "$\mathcal{F}$ is adequate" from "our attacks were weak," because they produced no signal even where signal exists. This one produces a large signal exactly where leakage is present and none where the certificate says there is none, which is the first measurement in this study that constrains, rather than merely names, the definitional gap the class restriction leaves open.

5.15 A paraphrase-aware score in $\mathcal{F}$

The limitation we have named most often is that every member of $\mathcal{F}$ reads the likelihood of the *literal* secret span, while Li et al. [9] show that gradient-ascent objectives do not delete probability mass so much as move it, onto semantically equivalent rephrasings. Mass that has migrated between surface forms is invisible to a score that reads one form, and averaging over query wrappers does not help: an average falls when mass moves out of the frames it averages over. We therefore implement the score that limitation calls for and add it to $\mathcal{F}$.

s_{para} takes the *maximum* token-normalised log-likelihood of the secret over five paraphrase frames of its carrier rather than the mean over wrappers, asking whether the secret survives under *any* rephrasing—the quantity an attacker free to rephrase

Table 20 A paraphrase-aware score added to the declared class, TOFU/GPT-2, 384 pairs, cohort target. $\hat{\Delta}$ is the realised cohort advantage under each score; "pairs to certify" is the stopping time under $\mathcal{F}$ and under $\mathcal{F} \cup \{s_{\text{para}}\}$. Enlarging the class costs power and no validity, and here the power costs one to two per cent.

	$\hat{\Delta}$		verdict, ε=0.2		pairs to certify	
subject	literal	paraphrase	$\mathcal{F}$	$\mathcal{F} \cup \{s_{\text{para}}\}$	$\mathcal{F}$	$\mathcal{F} \cup \{s_{\text{para}}\}$
no unlearning	+0.135	+0.141	**R**	**R**	384	384
GA	-0.016	-0.021	**I**	**I**	225	227
NPO	-0.057	-0.078	**I**	**I**	223	225
retrain (fresh adapter)	+0.089	+0.073	**I**	**I**	198	198

actually enjoys. It is a legitimate member of $\mathcal{F}$: it is a function of the deployed model's outputs on the committed pair, so Theorem 1 is untouched, and Theorem 2 needs no multiplicity correction to accommodate it.

Table 20 reports what happens when it joins the class, on TOFU/GPT-2 with the un-unlearned model, gradient ascent—the objective Li et al. [9] implicate directly—NPO, and retraining. Three things are worth stating, and the third is the one that matters.

First, the enlargement is nearly free, as the theory predicts: certification takes 227 pairs instead of 225 for gradient ascent and 225 instead of 223 for NPO, a cost of one to two per cent, and no verdict changes at any tolerance we tested. Second, s_{para} is not the binding constraint—its own e-process runs ahead of the literal one (log-wealth 17.0 against 15.8 for gradient ascent), so the certificate still waits on min-k%. Third, and this is the honest reading: s_{para} finds essentially the same advantage as the literal score everywhere, agreeing to within 0.02 in realised advantage on every subject.

We do not present that as evidence that the squeezing effect is absent, because this canary design cannot exhibit it. Our secrets are ten random alphanumeric characters, and a high-entropy random string has no semantically equivalent rephrasing for mass to move onto—only its carrier sentence does, which is what the frames vary. The experiment therefore establishes something narrower and still useful: the paraphrase-aware score is implemented, it costs almost nothing to carry, and it perturbs neither the verdicts nor the calibration. Testing the squeezing effect itself needs canaries whose *secret* is natural language and therefore paraphrasable, which is a different canary design—the twins would have to be exchangeable as *propositions* rather than as strings—and we name it as the next step rather than claiming this one covers it.

5.16 Can a provider find the cohort?

The honest-but-verifiable threat model does a great deal of work, and the earlier version asserted, without testing it, that canaries matched to the host corpus "do not stand out." They do. We measured it, and the result is unambiguous enough that we retract the claim: a perplexity filter under the *pristine base model*—a provider-side computation costing one forward pass per record and requiring no knowledge of the

Table 21 Canary detectability on TOFU under GPT-2 as the pristine base model (3,000 organic records, 384 canaries, one seed — this is a single configuration, not a sweep). "recall" is at a 1% false-positive rate on organic data. The predicted excess is $H \ln 2/T$; the observed excess exceeds it because the template's own phrasing is also off-distribution. Bold marks the one configuration a zero-knowledge filter substantially fails to recover.

			excess NLL/token		perplexity filter		detector
canary template	bits	tokens	predicted	observed	AUC	recall	recall
deployed (alphanumeric secret)	52	25	1.45	2.14	1.000	1.00	1.00
word-composed secret	42	35	0.84	1.80	1.000	1.00	1.00
entropy-diluted secret	68	88	0.54	0.87	0.974	**0.21**	1.00

template—separates the deployed canaries from the TOFU corpus with AUC 1.000 and recovers the entire cohort at a 1% false-positive rate (Table 21).

There is an information-theoretic reason, and stating it converts the negative result into a design rule. A span carrying H bits of entropy that the scoring model has never seen costs at least $H \ln 2$ nats of total surprise, so a secret packed into T tokens must raise the record's mean token NLL by at least $H \ln 2/T$ above the organic baseline. Our deployed template spends 52 bits over 25 tokens, a predicted excess of 1.45 nats per token against an organic standard deviation of 0.42: detection is not a failure of craft but an arithmetic certainty. Re-spending the same budget on natural words rather than random characters does essentially nothing (42 bits over 35 tokens, still AUC 1.000), confirming that the lever is T, not the vocabulary. Spending a *larger* budget over many more tokens does work: an entropy-diluted template carrying 68 bits over 88 tokens drops the perplexity filter's recall at 1% FPR from 1.00 to 0.21.

Two caveats keep this from being a clean fix. A *trained* detector—a character n-gram classifier fitted on labelled canaries—still recovers every cohort, diluted or not, because the template's fixed phrasing is a signature; but a provider who can fit such a classifier already knows the template, which is a far stronger position than the perplexity filter requires and one the threat model already excludes. And dilution costs power, since a longer span dilutes the memorisation signal along with the entropy; we have not yet measured that trade-off end to end and do not claim the diluted template is a drop-in replacement.

What detection does *not* do is defeat the audit, and the reason is worth stating plainly because it is the crux of the threat model. Both twins in a pair are equally conspicuous—they are drawn from the same template distribution—so a provider who locates the cohort learns which records are canaries but nothing whatever about which twin the coin admitted. The null of Theorem 1 survives complete disclosure of the manifest's *contents*; what it does not survive is a provider who, having located the cohort, treats it differently from organic data. That is special-casing, it is excluded by the honest-but-verifiable assumption, and it is exactly the gap cryptographic attestation [16] exists to close. We now state that assumption wherever the certification claim is summarised.

Table 22 Model axis, verdicts per seed for the core subjects at $\varepsilon = 0.2$ on the cohort target, spanning 0.9M to 5.1B parameters and release years 2019 to 2026. The seeds column states how many runs each cell aggregates; the 4 largest models have 1 seed each.

model	params	year	seeds	no unlearning	retrain	NPO
GPT-2	124M	2019	2	**R/R**	**I/I**	**I/I**
TinyGPT (ours)	0.9M	—	3	**R/R/R**	**I/I/I**	**I/I/I**
Qwen3-0.6B	0.6B	2025	1	**R**	**I**	**I**
Qwen3-4B-2507	4.0B	2025	1	**R**	**I**	**I**
Phi-4-mini	3.8B	2025	1	**R**	**I**	**I**
Gemma-4-E2B	5.1B	2026	1	**R**	**I**	**I**

5.17 A model axis from 2019 to 2026

To show that nothing about the framework is tied to a particular model generation, we run the pipeline across parameter scales and release years, from GPT-2 to the 2025–2026 cohort (Qwen-3 at 0.6B and 4B, Phi-4-mini, and the 5.1B-parameter Gemma-4-E2B), using a shared-base multi-adapter design that keeps a single frozen base in memory and represents every training stage as a LoRA adapter—a construction that fits 5B-parameter models on a 16 GB GPU and, incidentally, lets NPO's frozen reference be the fine-tuned adapter on the same base rather than a second model.

Table 22 collects the verdict patterns, now with the number of seeds stated per row, because the earlier version did not distinguish the three-seed rows from the single-seed ones and described the axis as "stable" on that basis. The corrected description is narrower. Detection is stable: every un-unlearned model is revoked at every scale, on every seed run. Certification is not uniformly stable: at the 384-pair cohorts we use for the largest models, the super-population target returns *undetermined* for retraining and NPO on Qwen3-0.6B and Qwen3-4B, and the single-seed rows cannot distinguish a systematic effect from noise. Under the cohort target those cells issue. We think the fair summary is that the framework's *detection* behaviour is scale-stable on the evidence we have, that its *certification* behaviour at the largest scales is limited by cohort size exactly as Theorem 4 predicts, and that one seed per cell is too thin to say more.

5.18 Ablations, guardrails, and cost

We close with the studies that probe the design choices and the practicality of the whole apparatus (Figure 8). On *betting strategy*, the online-Newton-step learner, the mixture, aGRAPA, and the truncated Krichevsky–Trofimov plug-in all certify within 20,000 pairs, whereas a fixed aggressive bet goes bankrupt and issues only 10% of the time—adaptivity is not a refinement but a requirement. On the *magnitude-aware arm*, sparse leakage in which only 5% of pairs are strongly memorised is detected in a median of 941 pairs against 2,196 for sign-only betting, a 2.3× speed-up. Proposition 7 says only that the magnitude arm is exact and more powerful under sparse leakage; the factor is measured, not predicted. A *guardrail on the protocol itself* confirms that planting canaries does not damage the model being audited: even at an extreme 38% canary share, held-out loss shifts by only 0.005 nats per character. We should be explicit that our own experiments sit far above a deployment share: the canary character share is 11.9% on TOFU and 22.5% in the tight-tolerance cohort of Section 5.11, against a target below 0.05% for a provider with millions of records. Academic corpora are simply too small to plant a statistically useful cohort at a deployment share, which is why the guardrail study exists; but it also means the detectability and dose–response numbers are measured at a share no deployment would use, and both would be harder to reproduce at 0.05%—detection because the flagged set's precision collapses, dose–response because the absolute cohort would have to be far larger.

On *cost*, the earlier version reported a single figure of 76 seconds without saying which configuration produced it, and compared it to alternatives without stating the hardware. Both are corrected here. Verification cost is $2 \cdot Q \cdot m$ short forward passes and no training at all: 1,536 passes per subject at the benchmark tier's $Q = 2$ and $m = 384$, and 5,120 at the GPT-2 synthetic tier's $Q = 4$ and $m = 640$. Measured end to end on the hardware each tier actually ran on, the median scoring time per certificate is 13 seconds for the small transformer, 63 seconds for GPT-2 at 640 pairs, 112–148 seconds for the 124M–1.4B benchmark tier, and 397–499 seconds for the 3.8B–5.1B models. The comparison to descriptive alternatives should be read as a ratio of *workloads*, not of wall-clock times on unmatched hardware: a retrain-based reference costs one full fine-tune of the same model on the same hardware, and a shadow-model evaluator at least 16, against VOUCH's zero. On our TOFU/GPT-2 configuration one fine-tune is 600 optimiser steps at batch 8, roughly 4,800 sequence-forward-and-backward passes, so a 16-model shadow panel is about 77,000 training passes against VOUCH's 1,536 inference passes, a ratio of about 50 to one—and the same ratio on any hardware, because both scale with the same model. A training pass is also several times more expensive than an inference pass, so the wall-clock gap is larger than 50×; we give the pass ratio because it is the part we can state without assuming a hardware model. We give a ratio rather than a speed-up factor because the two workloads are not interchangeable.

6 Limitations and conclusion

We set out to turn unlearning verification from a descriptive exercise into certification, and the ingredients that made it possible were an exact null created by the verifier's own randomness, a betting engine that survives continuous monitoring and streaming deletions, and a pair of composition rules that we proved must run in opposite directions. The empirical study shows the framework is not merely sound but practical: it calibrates correctly under adversarial stopping, matches or beats four valid sequential procedures at a fraction of their query cost, certifies exact unlearning on standard benchmarks across architectures at tolerances down to $\varepsilon = 0.05$, measures recoverability instead of arguing about it, and does so at a verifier cost of $2Qm$ forward passes and no training.

The claim a certificate makes is precise, and we regard stating its boundaries just as precisely as part of the contribution. VOUCH certifies the residual advantage

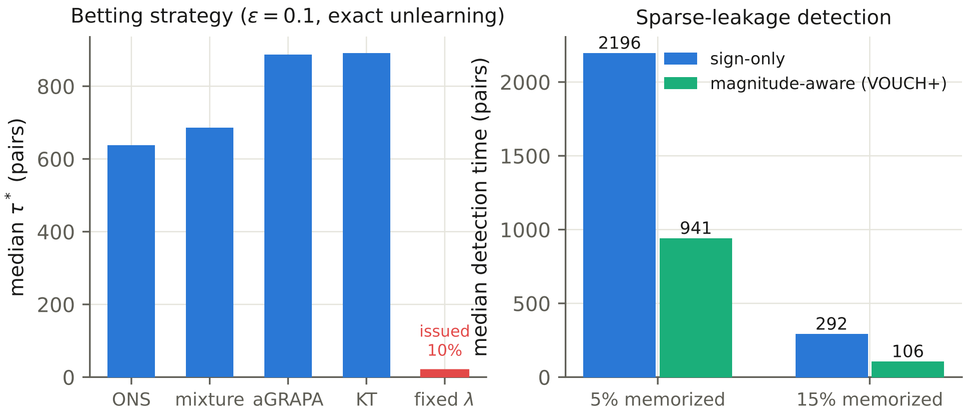

Fig. 8 Ablations. Left: betting strategies; a fixed aggressive bet goes bankrupt. Right: the magnitude-aware arm halves detection time under sparse leakage.

extractable through a *declared score class* $\mathcal{F}$, on a *committed canary cohort*, against the *deployed model*, under an *honest-but-verifiable* provider. Each of those four qualifications has a measured cost. Parameter-space erasure is behaviourally uncertifiable, so we do not claim it. The class restriction remains a definitional limit, but this version measures rather than merely names it. Two weak out-of-class attacks settle nothing, failing to beat the noise floor of subsampling. A LiRA-style shadow-model attack does better: with matched shadows it recovers an advantage of 0.82 on an un-unlearned model, and on a certified one returns -0.016 with a 95% interval that excludes every tolerance we certify at (Section 5.14). That is one tier, one target and 16 shadows, so it bounds the gap rather than closing it. Enlarging $\mathcal{F}$ costs power and no validity, and we now exercise that too: a paraphrase-aware score of the kind Li et al. [9] motivate is implemented and added to $\mathcal{F}$ (Section 5.15). The cohort statement transfers to the mechanism-level advantage only at the explicit inflation of Proposition 3, which is why our $\varepsilon = 0.05$ results are cohort certificates; the super-population arm reaches $\varepsilon = 0.1$ once the cohort clears 3,000 pairs and no further, exactly as Theorem 4 predicts. That experiment also exposed the sharpest limit we know of: at a single insertion per canary—the most organic-like dose—a 124M-parameter model memorises so little that the un-unlearned model certifies too, so the audit has no power in precisely the regime the canary-to-organic extrapolation needs it. The guarantee is per-cohort rather than per-user, since per-request certificates would require per-user canaries. The provider is assumed honest-but-verifiable, and we now know that assumption is easier to violate than we had supposed: a perplexity filter costing one forward pass per record locates the entire planted cohort, so a provider willing to special-case it can find it, and closing that gap needs cryptographic attestation that VOUCH does not itself supply. The audit's own randomness is now as checkable as its commitment—the reveal order can be drawn from a public randomness beacon bound to the manifest root, which is what Theorem 2's hypothesis actually requires—though the results reported here predate that and were computed with a seed-derived order. Canaries must be planted before fine-tuning, so legacy models cannot be retro-certified. Both two-sided certification and consensus-based streaming composition cost power, which we quantify (Tables 6 and 8) so that cohorts can be sized accordingly. And the extrapolation from planted canaries to organic records remains calibrated by the dose–response strata rather than proven—with the $r = 1$ stratum, the one most like an organic record, carrying the weakest signal.

Within those boundaries, the framework delivers what a deletion audit actually needs: an error-controlled, continuously monitorable, black-box certificate of bounded residual influence on a committed cohort. The released harness reproduces the entire study on free, ephemeral compute, which we hope lowers the barrier to auditing unlearning wherever it is deployed.

Appendix A Reproducibility

Everything below is fixed by the released code and its committed seeds; this appendix collects the settings a reader would otherwise have to recover from source.

What "retraining from scratch" means here.

All fine-tuning and all unlearning in this paper operate on LoRA adapters of rank 16 ($\alpha_{\mathrm{LoRA}} = 32$, dropout 0, target modules `all-linear`) over a *frozen* pretrained base model. "Retrain" therefore means: initialise a *fresh* adapter on the same frozen base and train it on D_{keep} only—the keep split with neither the organic forget data nor any canary—using the identical optimiser, schedule and step count as the original fine-tune. It is exact unlearning *within the adapter paradigm*: the adapter that saw the canaries is discarded and never contributes to the retrained model, and the frozen base never saw a canary at any point. It is not retraining of the base model's pretraining, which we never perform and do not claim. We now call this subject "retrain (fresh adapter)" in the tables to remove the ambiguity.

Whether adapter-level unlearning is representative of a deployment setting deserves a direct answer, since the certificate is meant to cover the latter. The audit itself is indifferent: it queries M_u as a black box and never inspects parameters, so nothing in Theorem 1 or Theorem 2 changes if the provider unlearns by full-parameter gradient steps instead. What the adapter setting does change is the *difficulty* of the unlearning problem the subjects face. A rank-16 adapter has far fewer degrees of freedom in which to hide memorised content than a full fine-tune, so our subjects are, if anything, being certified in a regime that favours them; a provider doing full-parameter unlearning should expect verdicts at least as demanding. We flag this as a limitation of the empirical study rather than of the framework, and full-parameter runs are the natural next scaling step.

Models.

`gpt2` (124M), `EleutherAI/pythia-160m` (160M), `microsoft/phi-1_5` (1.4B), `Qwen/Qwen3-0.6B`, `Qwen/Qwen3-4B-2507`, `microsoft/Phi-4-mini-instruct`, `google/gemma-4-E2B-it`, and an in-repo 0.9M-parameter TinyGPT for the CPU-affordable tier. Phi-1.5 LoRA fine-tuning diverges to NaN in fp16, so all Phi-1.5 runs are fp32; the remaining large models run fp16 with a shared frozen base and one adapter per training stage.

Data.

TOFU [12]: keep = `retain90` minus a 200-record held-out slice, forget = `forget10` (400 QA pairs), public probe set for P1 and the capability reading = `world_facts`, utility = the held-out retain slice, all formatted `Question: {q}\nAnswer: {a}`. MUSE-News [13] raw split: keep = 3,000 500-character chunks of `retain1`, forget = 500 chunks of `forget`, public/holdout = 600 chunks of `holdout` split 400/200 into the relearn corpus and the utility set.

Canaries.

Templated generator, deterministic in the run seed; domains `qa` for TOFU and {`pii`, `fact`} for MUSE-News; repetition strata $r_i \in \{1, 2, 4, 8\}$ assigned round-robin; secrets of 10 alphanumeric characters (51.7 bits). Cohort sizes: 384 pairs (TOFU/GPT-2, TOFU/Pythia, all model-axis runs), 512 (TOFU/Phi-1.5, all MUSE-News runs, TinyGPT), 640 (the GPT-2 synthetic tier). Canary character share of the fine-tuning corpus is 11.9% on TOFU and reported per run in the released JSON; the guardrail study (Section 5.18) covers shares up to 38%.

Training and unlearning.

Fine-tune and retrain: 600 steps, batch 8, AdamW at learning rate 2×10^{-4}, block length 160, gradient-norm clip 1.0. Unlearning stages use batch 4 and learning rate 1×10^{-4} (half the fine-tune values) with the same block and clip: gradient ascent 100 steps on the forget set; GradDiff 250 steps, ascent on forget plus descent on keep with unit weight; NPO 250 steps with $\beta = 0.1$ and the fine-tuned adapter as the frozen reference; SimNPO 250 steps with $\beta = 0.1$ and margin $\gamma = 0$, reference-free on the length-normalised forget NLL; "NPO (25% budget)" truncates NPO to 62 steps. Probes: P1 relearn = 100 steps on the public corpus, which never contains a canary; P2 = symmetric per-channel round-to-nearest 4-bit quantisation of all linear and embedding weights; P3 = three adversarial prompt wrappers applied at scoring time only. Probe budgets are fixed across all subjects and models.

Scoring.

$Q = 2$ query wrappers for every benchmark and model-axis run (identity and "According to the archives, ..."); $Q = 4$ for the GPT-2 synthetic tier, which additionally reports the base-calibrated `ratio` score. Score class $\mathcal{F} = \{s_{\text{loss}}, s_{\text{mink}}\}$ for the benchmark and model-axis runs and $\mathcal{F} = \{s_{\text{loss}}, s_{\text{mink}}, s_{\text{ratio}}\}$ for the synthetic tier, with $k = 20\%$ for min-k. Scores are computed from temperature-0 forward passes and aggregated over wrappers by the mean. Verification cost is exactly $2Qm$ forward passes per subject.

Two implementation notes that affect validity.

First, the stake cap and the wealth factor must be computed from the *same* clamped value of the recentred boundary $p_0(t)$. The admissible range is $\lambda_t < c/(1 - p_0(t))$, so deriving the cap from one clamped boundary and the payoff from another lets the worst-case factor $1 + \lambda_t(p_0(t) - 1)$ fall below zero. The case is reachable exactly when $m\,p_0$ is an integer—which holds for $m = 640$ at every tolerance we report, in both directions—so it is not a corner a reimplementation can ignore. The released code derives both from one clamp and carries a regression test for it.

Second, Theorem 2 requires the reveal order π to be independent of the realised sign vectors. Every result table in this paper was produced with `reveal_source = "seeded"`, in which π is seeded from the run seed—the same seed that drives cohort generation and training—so for those runs the independence does not strictly hold, even though nothing in the pipeline exploits the coupling. We report them as they were computed rather than restating them under a different reveal rule. The released code now also implements the two sources that do satisfy the hypothesis (`vouch/verify/beacon.py`): `"beacon"` seeds π with $H(\text{manifest root} \,\|\, r)$ for the randomness r of a drand round drawn after the commitment is published, recording the round number and value on the certificate so a third party can recompute the order; and `"local"` draws from operating-system entropy. A deployed audit should use the beacon, and the certificate object now carries a `reveal` block naming the source, whether it is third-party verifiable, and whether it meets Theorem 2's hypothesis.

Verification.

$\alpha = 0.05$; two-sided certification; sign-based revocation for all headline tables, with the magnitude-aware arm reported separately; finite-cohort certificate target by default and the super-population target reported alongside wherever tolerance matters. Confidence-sequence grid 1001 points on $[0, 1]$. Betting mixture: fixed fractions $\{0.02, 0.05, 0.1, 0.2, 0.4, 0.7\}$ of $\lambda_{\max}$, an ONS expert with $\eta = \frac{1}{2}$ and $A_0 = 1$, and a truncated KT plug-in, combined with uniform prior weights. Reveal order: a permutation seeded by the run seed, committed before any query. Tie-breaking: a committed PRNG seeded at 20260702, shared across every run so that tie decisions are reproducible. Early stopping is enabled for the certificate arm; the revocation arm continues to accumulate evidence after a crossing so that the streaming product alarm is well defined.

Simulation tiers.

Validity, soundness, dependence and baselines: 2,000 seeds. Power and censoring: 500 seeds per cell with a 20,000-pair horizon. Tightness and streaming: 500 seeds. Group-sequential boundaries use $K = 10$ equally spaced looks with Pocock and O'Brien–Fleming spending functions, calibrated by exact binomial dynamic programming over the partial-sum lattice at the least favourable null.

Hardware and cost accounting.

The study ran on heterogeneous free and local compute: Colab and Kaggle T4/P100 GPU sessions for the 1.4B–5.1B models, an Apple-silicon laptop (MPS backend) for the GPT-2 and Pythia benchmark runs and the revision-round SimNPO runs, and Colab CPU sessions plus a four-core container for the TinyGPT tier and all simulations. Because the hardware is not matched across tiers, we report verifier cost as a *workload* in forward passes and give measured wall-clock times only alongside the tier that produced them (Section 5.18). The comparison against retrain-based and shadow-model evaluators is likewise stated as a ratio of workloads on the same model and hardware—1,536 inference passes against roughly 4,800 training passes per fine-tune and 16 fine-tunes for a shadow panel—rather than as a wall-clock speed-up factor.

Data availability statement

The complete source code of the VOUCH framework, every experimental result reported in this paper (including all per-seed JSON records, generated tables, and figures), and the fault-tolerant execution harness that produced them are publicly available in an open repository; the repository reference is withheld here to preserve double-blind review and is provided to the editorial office in the cover letter. The TOFU and MUSE-News benchmark datasets used in the experiments are publicly available from their original releases [12, 13]; all canary data are generated deterministically from committed seeds by the released code.

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